

NYANSAPO — GROUP 1

Complete Supervisor Feedback Archive

This document merges all supervisor feedback received to date on the Nyansapo project, in chronological order, for direct comparison against the current revision. Each report is reproduced in full and unedited.

#	Report	Date	Focus
1	Architecture Audit	May 2026	Initial system architecture review
2	Dataset Size Advisory	May 2026	Dataset scale and sufficiency
3	Revised Assessment	May 2026	Post-revision proposal assessment
4	Draft 0 Experiment Feedback	Jul 2026	8 fatal/major flags — self-referential scoring, test leakage, RAGAS misla
5	Verification Step 2	Sep 2026	57.4% compliance — C1/F8 confirmed open via closeout audit script
6	Verification Checklist	Sep 2026	ESCALATE verdict — C4, X1–X7 code-level defects (arity break, leaked

Current status as of this archive: reports 1–3 fully closed. Report 4's 8 flags fully addressed. Report 5's C1 and F8 confirmed closed via the v3 notebook rebuild (vocabulary-set self-test, single canonical train_model_v2). Report 6's X1–X6 code defects resolved in v3; C6 (post-hoc deployment seed selection) disclosed as a permanent, code-verified limitation.

MODEL & ARCHITECTURE AUDIT

Are the Nyansapo stack components fit for Tier 1 publication?

A solution-based technical advisory for Dr. Eric Opoku Osei and Team Nyansapo

Prepared for	Dr. Eric Opoku Osei · Team Nyansapo
Research project	Nyansapo — RAG-Grounded Tutoring for Localized KNUST Multimedia Curricula
Review date	March 2026
Assessment standard	Tier 1 & 2 Journal Publication (Computers & Education: AI · Frontiers in AI)
Verdict summary	3 critical upgrades required · 2 components are novel · 1 is genuinely market-leading

SECTION 1 — ARCHITECTURE OVERVIEW AUDIT

The table below maps every component in the Nyansapo stack against the 2025–2026 Tier 1 editorial standard. Each component is rated and given a specific upgrade recommendation.

COMPONENT	CURRENT CHOICE	VERDICT	RECOMMENDED UPGRADE	URGENCY FOR TIER 1
Generator LLM	Mistral-7B-Instruct v0.2 (Sept 2023)	MARGINAL	Mistral Small 3 (2025) or keep v0.2 + full justification	MEDIUM
Embedding model	all-MiniLM-L6-v2 384-dim · 2021	OUTDATED	BGE-M3 (BAAI, 2024) 1024-dim · academic	HIGH
Vector index	FAISS Flat (exact) no reranking	ACCEPTABLE	FAISS HNSW + cross-encoder reranker (BGE reranker-base)	MEDIUM
Retrieval strategy	Dense (FAISS) only BM25 as baseline k=3 flat	WEAK	Hybrid BM25+dense fused by Reciprocal Rank Fusion (RRF)	HIGH
RAG variant	Naive RAG (Lewis 2020) no self-correction	DATED	Corrective RAG (CRAG, Yan 2024) lightweight evaluator	CRITICAL
Evaluation framework	BERTScore+ROUGE + human trust No faithfulness	PARTIAL	Add RAGAS: Faithfulness + Answer + Context Relevance	HIGH
Chunking strategy	Recursive char splitter, 300 tok no metadata	ACCEPTABLE	Contextual retrieval: add topic/lecture metadata per chunk	LOW
Offline mode	Ollama + Mistral-7B Enhancement A (planned)	NOVEL	Use Phi-4-mini (3.8B) instead — runs on 8GB RAM, no GPU	KEEP + UPGRADE
Orchestration	LangChain 0.1.12	DATED	LangChain 0.3+	LOW

COMPONENT	CURRENT CHOICE	VERDICT	RECOMMENDED UPGRADE	URGENCY FOR TIER 1
	FastAPI + React Native		or LangGraph pipeline FastAPI/RN: keep	

VERDICT KEY: **NOVEL** = unique, no prior paper addresses this. **ACCEPTABLE** = functional, reviewers will not object. **MARGINAL** = usable but will attract reviewer comments. **OUTDATED / WEAK / DATED** = will cause desk rejection without correction.

SECTION 2 — THE FIVE UPGRADES THAT MATTER MOST

The following five components require specific action before the manuscript is submitted. They are ordered by impact on acceptance probability, not by effort. Read each section carefully — the fixes are concrete and achievable within the existing experiment timeline.

UPGRADE 1 (CRITICAL) — Replace Naive RAG with Corrective RAG (CRAG)

The team is implementing the Lewis et al. (2020) naive RAG pipeline: retrieve → concatenate → generate. Every Tier 1 reviewer in 2025–2026 knows that naive RAG has a well-documented failure mode: when retrieval quality is poor, the wrong chunks are still passed to the generator, and hallucinations occur regardless of the retrieval step. Submitting a paper about hallucination reduction that uses a 2020-era pipeline without addressing this known failure mode will receive the standard comment: 'The authors do not address Corrective RAG or Self-RAG, which are the current standard for robust hallucination mitigation.'

CRITICAL — This is the single most likely cause of desk rejection.

What is the problem?

Corrective RAG (CRAG, Yan et al., 2024) is a drop-in enhancement to the Lewis et al. (2020) pipeline. A lightweight retrieval evaluator (fine-tuned T5-small, 0.77B parameters) is inserted after the retrieval step. It scores the quality of the retrieved chunks. If the score is low (poor retrieval), it triggers a corrective action: the query is rewritten using a knowledge refinement module and re-retrieval is attempted before the generator is called. The original pipeline is unchanged for high-quality retrievals.

Exactly what to do:

- Add the CRAG evaluator as a post-retrieval quality check in the pipeline.
- If evaluator score < threshold (tune this on the validation set), trigger query rewriting (using Mistral-7B to rephrase the student query) and re-retrieve once.
- Pass the corrected chunks to Mistral-7B-Instruct for generation as normal.
- Report the CRAG evaluator score distribution across the 5,000 test queries as a new result — this is a contribution in itself.
- Citation to add: Yan, S., Gu, J., Zhu, Y., & Ling, Z. (2024). Corrective Retrieval Augmented Generation. arXiv:2401.15884. <https://doi.org/10.48550/arXiv.2401.15884>

Effort estimate:

3–5 additional implementation days. The T5-small evaluator checkpoint is available on HuggingFace (the CRAG paper authors released it). This is a pip install + model load, not a full system redesign.

Impact on acceptance probability:

Changes the paper from 'we applied RAG' (2020 architecture) to 'we applied corrective RAG with an automatic retrieval quality evaluator' (2024 architecture). Directly on-trend for every major EdTech and HCAI journal in 2025–2026.

UPGRADE 2 (HIGH) — Replace the Embedding Model: all-MiniLM-L6-v2 → BGE-M3

all-MiniLM-L6-v2 is a 2021 general-purpose sentence embedding model with 384 dimensions. It was trained on general English web text. Academic lecture transcripts — with course-specific terminology, equation descriptions, and domain jargon specific to Computer Science at KNUST — are exactly the kind of specialised text where this model degrades in retrieval precision. A Tier 1 reviewer with ML expertise will flag this immediately.

HIGH — This is a one-line code change with significant impact on retrieval quality.

What is the problem?

BGE-M3 (BAAI General Embedding, 2024) is a multilingual, multi-task dense retrieval model with 1024 dimensions. It was specifically designed and benchmarked for document retrieval and RAG pipelines. It outperforms all-MiniLM-L6-v2 by a substantial margin on academic text retrieval benchmarks. It is open-weights, free, and runs on the same Colab A100 hardware.

Exactly what to do:

- Replace: from sentence_transformers import SentenceTransformer
- Old: model = SentenceTransformer('all-MiniLM-L6-v2')
- New: model = SentenceTransformer('BAAI/bge-m3')
- Update FAISS index dimension from 384 to 1024.
- Rebuild the FAISS index — this takes approximately 15 additional minutes.
- Citation to add: Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., & Liu, Z. (2024). BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. arXiv:2402.03216.

Effort estimate:

30 minutes of code change. 15 minutes of re-indexing. Run a quick ablation comparing MiniLM vs BGE-M3 retrieval precision on 50 sample queries — report this in the ablation study.

Impact on acceptance probability:

Improves retrieval precision, which directly improves BERTScore (the primary metric). Demonstrates current awareness of the embedding model landscape to reviewers. Directly supports the paper's hallucination-reduction claim.

UPGRADE 3 (HIGH) — Add Hybrid Retrieval with Reciprocal Rank Fusion (RRF)

The current pipeline uses FAISS dense retrieval only, with BM25 as a comparison baseline. This is backwards from the 2024–2025 standard. BM25 should not be a baseline that RAG is compared against — it should be fused WITH dense retrieval to produce a stronger hybrid retrieval signal. Dense retrieval excels at semantic matching ('explain database relationships') but is weak on exact-match queries ('what is the definition of third normal form in Week 4'). Student queries in a real university setting contain both types. Using only dense retrieval means the system will systematically fail on terminology-exact queries.

HIGH — Purely dense retrieval is known to miss exact-match queries that BM25 handles well.

What is the problem?

Hybrid retrieval combines BM25 (keyword-based) and dense (semantic) retrieval scores. Reciprocal Rank Fusion (RRF) is the standard fusion method: each document's rank from both retrievers is converted to a score ($1 / (k + \text{rank})$), and the scores are summed. The top-k documents by combined RRF score are passed to the generator. This is the retrieval strategy used in virtually all 2024–2025 top-performing RAG systems.

Exactly what to do:

- Use LangChain's EnsembleRetriever: combine BM25Retriever and FAISSRetriever with weights=[0.4, 0.6].
- Or implement RRF manually: retrieve top-20 from each, compute RRF scores, select top-3.
- Run ablation: BM25-only vs Dense-only vs Hybrid. This ablation table is itself a publishable contribution.
- Citation to add: Cormack, G. V., Clarke, C. L., & Buettcher, S. (2009). Reciprocal rank fusion outperforms condorcet and individual rank learning methods. SIGIR 2009.

Effort estimate:

1–2 days implementation. The ablation comparing all three retrieval strategies (BM25 only, dense only, hybrid RRF) is the correct ablation for a retrieval paper and replaces the current weaker ablation plan.

Impact on acceptance probability:

Makes the retrieval component publication-standard for 2024–2025. Hybrid retrieval is now the expected default — reviewers will ask why it was not used if absent.

UPGRADE 4 (HIGH) — Add RAGAS Evaluation Suite

The team's current metrics — BERTScore + ROUGE-L + human trust study — are good but incomplete for a paper whose central claim is hallucination reduction. BERTScore measures semantic similarity to a reference answer. It does not measure whether the generated answer is actually grounded in the retrieved chunks. A model can achieve a high BERTScore by hallucinating fluently if the hallucinated answer happens to overlap semantically with the gold-standard answer. This is exactly the failure mode the paper claims to fix, and reviewers know it.

HIGH — BERTScore alone is insufficient to prove hallucination reduction for Tier 1.

What is the problem?

RAGAS (Retrieval Augmented Generation Assessment) is a pip-installable Python library that provides three metrics that directly address the hallucination reduction claim: (1) Faithfulness — is the answer derivable from the retrieved context? This is the core anti-hallucination metric. (2) Answer Relevance —

does the answer address the actual question? (3) Context Relevance — did the retriever find relevant chunks? These three metrics together tell a complete story about RAG quality that BERTScore + ROUGE alone cannot tell.

Exactly what to do:

- Install: `pip install ragas`
- After generating answers for all 5,000 test queries, run:
- `from ragas import evaluate`
- `from ragas.metrics import faithfulness, answer_relevancy, context_relevancy`
- `results = evaluate(dataset, metrics=[faithfulness, answer_relevancy, context_relevancy])`
- Report RAGAS scores for RAG vs Vanilla LLM baseline — the Faithfulness score for the Vanilla LLM (no retrieval) should be near zero, confirming your hallucination reduction claim quantitatively.
- Citation: Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2023). RAGAS: Automated evaluation of retrieval augmented generation. arXiv:2309.15217.

Effort estimate:

1 day to integrate. RAGAS requires an LLM-as-judge under the hood (it calls the generator model to evaluate outputs) — this adds approximately 2–3 hours to total evaluation runtime on Colab Pro+.

Impact on acceptance probability:

RAGAS Faithfulness is the single most direct quantitative proof of the paper's hallucination reduction claim. Without it, reviewers can argue that BERTScore improvements do not prove grounding. With it, the paper has an unambiguous anti-hallucination metric. This is now expected at Computers & Education: AI and Frontiers in AI for RAG evaluation papers.

UPGRADE 5 (MEDIUM) — Generator LLM: Justify Mistral-7B-Instruct-v0.2 or Upgrade

Mistral-7B-Instruct-v0.2 was released in September 2023. It is now three model generations behind (Mistral Small 3 was released in 2025). It is not dead — it is still a capable, well-benchmarked model — but a Tier 1 reviewer will immediately ask why the team did not use a more recent model. The team must make a deliberate, cited choice.

MEDIUM — Must be addressed explicitly in the manuscript, one way or the other.

What is the problem?

There are two defensible paths. Path A: Upgrade to Mistral Small 3 (2025). Same parameter count, same Colab hardware requirements, significantly better instruction-following on MT-Bench 2024. Requires a one-line model name change and re-running experiments. Path B: Keep Mistral-7B-Instruct-v0.2 and explicitly justify it. The justification must be: this is the most capable model proven to run on the target resource-constrained hardware (Colab A100 AND offline laptop without GPU via Ollama), enabling direct comparison between the online and offline deployment modes. This is a legitimate HCAI justification that a reviewer will accept IF it is stated explicitly and cited.

Exactly what to do:

- For the offline mode specifically: use Phi-4-mini (3.8B parameters, Microsoft 2025) instead of Mistral-7B. Phi-4-mini runs on 8GB RAM without any GPU — this means it works on any KNUST student's laptop. This is the correct model for the resource-constrained offline use case.
- The paper then compares three systems: Online RAG (Mistral-7B/Small3 + FAISS server), Offline RAG (Phi-4-mini + local FAISS), and Vanilla LLM baseline (Mistral-7B, no retrieval).
- Citation for Phi-4: Abdin, M., et al. (2024). Phi-4 Technical Report. arXiv:2412.08905.

Effort estimate:

If upgrading: 1 day to swap model and re-run experiments. If justifying: 2 paragraphs added to Section 4.1 of the manuscript.

Impact on acceptance probability:

Prevents reviewer comment on model currency. Using Phi-4-mini for offline mode adds a genuine second model contribution and makes the offline comparison more compelling — a 3.8B model running without internet vs a 7B model requiring server infrastructure is a strong story for African EdTech deployment.

SECTION 3 — WHAT IS GENUINELY NOVEL IN THE NYANSAPO STACK

Amid the components requiring upgrade, two elements of the Nyansapo architecture are genuinely novel and must be protected and strengthened — not changed.

Novel Component 1 — Offline / Low-Bandwidth RAG Inference Mode

VERDICT: GENUINELY NOVEL — No prior RAG-in-education paper addresses this.

Every published RAG-in-education paper assumes stable internet connectivity and cloud API access. Nyansapo plans an offline inference mode using a locally running LLM, directly addressing the bandwidth-constrained reality of sub-Saharan African universities where connectivity is intermittent and expensive. This is not incremental — it is the first RAG-in-education paper to evaluate this deployment scenario.

The recommended model for offline mode is Phi-4-mini (3.8B, Microsoft 2025). At 3.8B parameters with 4-bit quantisation, Phi-4-mini runs on a standard laptop with 8GB RAM and no GPU — making it viable on any KNUST student device. The paper then reports a direct comparison of answer quality and latency between online (Mistral + server) and offline (Phi-4-mini + local FAISS) modes. This two-mode evaluation is a unique contribution that no prior paper has made.

Do not remove or downgrade this component. It is the primary differentiator that positions the paper above all existing RAG-in-education work.

Novel Component 2 — First RAG Deployment Study in Sub-Saharan African Higher Education

VERDICT: NOVEL POPULATION — Confirmed gap by Gao et al. (2023) systematic review.

The systematic review by Gao et al. (2023) — already cited in the proposal — explicitly confirms no prior RAG study in sub-Saharan African public university settings. The KNUST institutional corpus, Ghanaian student population, and resource-constrained infrastructure together constitute a population novelty that is verifiable, citable, and directly relevant to the SDG 4 and SDG 10 framing.

This novelty is strengthened — not weakened — by the CRAG, hybrid retrieval, and RAGAS upgrades described above. A paper that deploys a state-of-the-art 2024–2025 RAG architecture in an African university for the first time is a significantly stronger contribution than one that deploys a 2020-era naive RAG pipeline in the same context.

SECTION 4 — PRIORITISED ACTION LIST FOR EXPERIMENT PHASE (April–June 2026)

The table below maps every architecture upgrade to the specific experiment phase in the current timeline. All upgrades are compatible with the Colab Pro+ environment and the May–June experiment window.

#	ACTION	EFFORT	IMPACT ON ACCEPTANCE	DEADLINE
1	Add Corrective RAG (CRAG) evaluator	3–5 days	Eliminates most common desk-rejection reason for RAG papers in 2025–2026. Upgrades architecture from 2020 to 2024 standard.	Before experiment run (May 11–31)
2	Replace embedding model: all-MiniLM-L6-v2 → BGE-M3	30 min code + 15 min re-index	Improves retrieval precision on academic text. Prevents reviewer flag on outdated embedding model.	Before FAISS index build (May 6–10)
3	Add hybrid BM25 + dense retrieval with RRF fusion	1–2 days	Closes the known dense-only retrieval gap on exact-match queries. Turns BM25 from baseline to system component.	During RAG pipeline build (May 11–20)
4	Add RAGAS evaluation suite (Faithfulness + Answer Relevance + Context Relevance)	1 day + 2–3 hr runtime	Provides direct quantitative proof of hallucination reduction. Expected at Computers & Education: AI for RAG evaluation papers.	Evaluation phase (Jun 1–10)
5	Upgrade LangChain 0.1.12 → LangChain 0.3+ in requirements.txt	1 hour	Corrects the reproducibility checklist. Prevents reviewer comment on outdated orchestration library.	Environment setup (Mar–Apr)
6	Use Phi-4-mini (3.8B) for offline mode instead of full Mistral-7B	1 day	Phi-4-mini runs on 8GB laptop RAM without GPU — genuinely deployable at KNUST without server infrastructure. Strengthens the Africa-specific deployment story.	Offline mode implementation (May 11–31)
7	Add contextual metadata (topic + lecture week) to each FAISS chunk	10-line Python change	Improves retrieval recall on topic-specific queries. Low effort, meaningful gain.	Preprocessing phase (Apr 15–30)

SECTION 5 — NOTE TO DR. ERIC OPOKU OSEI

This architecture audit confirms that the Nyansapo team's fundamental research direction is sound and the core novelty — offline RAG for African university deployment — is real and unoccupied in the literature. The issues identified are not structural flaws; they are component-level upgrades that bring a good proposal to Tier 1 publication standard.

The most important message for the team is this: the difference between a paper that gets desk-rejected and a paper that gets reviewed at Computers & Education: AI is not the quality of the experiment — it is whether the architecture uses 2024–2025 methods or 2020–2022 methods. The CRAG upgrade, the BGE-M3 embedding swap, and the RAGAS evaluation addition together require approximately one additional week of implementation work. They transform the paper from 'we applied RAG' to 'we applied corrective hybrid RAG with a retrieval quality evaluator, evaluated using RAGAS faithfulness metrics, in the first offline-capable deployment study in sub-Saharan African higher education.' These are two very different papers — and only one of them passes the Tier 1 editorial filter.

Checklist for Dr. Osei — before approving the manuscript draft

- CRAG evaluator (Yan et al., 2024) is implemented and its retrieval quality score distribution is reported.
- Embedding model is BGE-M3, not all-MiniLM-L6-v2. FAISS index dimension is 1024.
- Retrieval is hybrid (BM25 + dense, fused by RRF), not dense-only.
- RAGAS Faithfulness score is reported for RAG vs Vanilla LLM baseline.
- Offline mode uses Phi-4-mini (3.8B), not Mistral-7B. Runs on 8GB laptop RAM confirmed.
- LangChain version in requirements.txt is 0.3+ not 0.1.12.
- Ablation study covers: CRAG vs Naive RAG, Hybrid vs Dense-only, BGE-M3 vs MiniLM.
- Related Work section cites CRAG (2024), BGE-M3 (2024), RAGAS (2023), Hybrid RRF (2009).

A final word to Team Nyansapo

The architecture upgrades in this document might look like a long list, but most of them are one-line code changes or pip installs. The hard work — the KNUST corpus, the user trust study, the offline deployment mode, the ethics clearance, the 5,000-query evaluation — is already planned and is genuinely novel. What this document is asking you to do is spend one additional week making sure the scaffolding around that hard work is built to the 2025 standard, not the 2020 standard. That one week is the difference between a paper that reviewers read and one that gets returned at the desk. You have already done the harder thing. Do the easier thing too.

DATASET SIZE REQUIREMENTS FOR TIER 1 PUBLICATION

A Study-Specific Advisory for the Nyansapo RAG Project

Minimum · Average · Maximum · Recommended

1. Purpose and Scope of This Advisory

This advisory analyses the dataset size requirements that must be met for the Nyansapo RAG project to produce claims that are defensible at Tier 1 journal level — specifically Computers & Education: AI (Elsevier) and Frontiers in AI. The analysis is grounded exclusively in the work as designed: a Retrieval-Augmented Generation (RAG) system using an institutional corpus of KNUST Computer Science multimedia lecture transcripts, evaluated against a human-annotated gold-standard question-answer benchmark, with a synthetic data augmentation layer for student interaction logs.

The figures cited here are drawn from the published norms of comparable RAG evaluation studies in educational NLP (2022–2025), not from generic ML dataset standards. Overstating dataset requirements for this class of experiment is a common error that leads students to scope-creep, delays, or — worse — fabricating data to meet phantom minimums. This document avoids that error by studying what peer-reviewed papers in this exact sub-field actually used.

2. Nature of the Nyansapo Experiment

Before establishing dataset size thresholds, the nature of the experiment must be clearly characterised, as reviewers calibrate their expectations accordingly. The Nyansapo study is:

- **A retrieval evaluation study** — measuring whether RAG retrieval reduces hallucination in LLM-generated answers relative to a no-retrieval baseline.
- **An institutional corpus study** — the knowledge base is not a public benchmark; it is a domain-specific, institution-specific collection of lecture transcripts. This is a strength, not a weakness.
- **A human-centred AI (HCAI) evaluation** — user trust and cognitive load (NASA-TLX) are primary outcome variables alongside automated metrics.
- **NOT a model training study** — Mistral-7B-Instruct is used off-the-shelf with no fine-tuning. This is critical: dataset size requirements for training are entirely different from those for evaluation. The team is conducting an evaluation study.

CRITICAL FRAMING: Because Nyansapo does not train a model, the dataset size requirements

are governed by evaluation adequacy and statistical power, not training data volume. Tier 1 RAG evaluation studies with 2,000–10,000 test queries are common and well-regarded. Requirements of 100,000+ items belong to pre-training studies — a completely different class of work.

3. The Three Dataset Components and Their Size Requirements

The Nyansapo dataset has three distinct components, each with its own size logic. Conflating them produces incorrect requirements. Each is addressed separately below.

3.1 Component A: The Retrieval Corpus (Lecture Transcripts)

The retrieval corpus is the knowledge base from which the RAG system retrieves chunks. Its adequacy is measured by coverage and chunking density, not raw document count.

THRESHOLD	CORPUS SIZE	FAISS CHUNKS (300 tok)	RATIONALE
Minimum	50 transcripts ~375,000 words ~500,000 tokens	~1,650 chunks	Covers ~1 academic semester of content. Sufficient for a proof-of-concept RAG study. Comparable to baseline studies reviewed in TutorBot (2023) and EduRAG (2024).
Average (Peer Norm)	100 transcripts ~750,000 words ~1,000,000 tokens	~3,300 chunks	Covers 2 semesters across multiple CS modules. This is the median corpus size in RAG-for-education papers published 2023–2025 in SSCI-indexed journals.
Maximum (Tier 1 Strong)	200+ transcripts ~1,500,000 words ~2,000,000 tokens	~6,600 chunks	Covers the full Computer Science curriculum at KNUST. Maximum academic coverage. FAISS flat index at this size runs on Colab A100 in under 3 minutes. No hardware constraint.
Nyansapo Plan	150 transcripts ~1,125,000 words ~1,500,000 tokens	~4,950 chunks	ABOVE the peer average. Well within Tier 1 acceptability. The Multimedia Unit request (Maxwell, March 18–20) directly targets this corpus. This plan is correct and should not be increased.

RECOMMENDATION: The team's plan of 150 transcripts is the correct target. It exceeds the peer average, is feasible within the institutional access request already in motion, and produces a FAISS index of ~5,000 chunks that runs efficiently on Colab Pro+ (A100). Do not inflate this

target.

3.2 Component B: The Gold-Standard Evaluation Set (Human-Annotated QA Pairs)

This is the most scrutinised dataset component in any RAG evaluation paper. Reviewers examine: (1) how many annotated pairs exist, (2) who annotated them and with what inter-rater protocol, and (3) whether the test set is large enough to support statistical significance claims.

The team has planned 200 human-annotated question-answer pairs plus LLM-generated synthetic augmentation to 5,000 test queries. This two-tier structure is methodologically sound and reflects current practice. The figures below analyse both tiers.

Tier 1 of the Evaluation Set: Human-Annotated Gold Standard

THRESHOLD	QA PAIRS	STATISTICAL BASIS	RATIONALE
Minimum	80–100 pairs	Wilcoxon signed-rank, n=80 achieves power=0.80 at medium effect (d=0.5)	Seen in pilot studies and workshop papers. Accepted in conference venues. For Computers & Education: AI, this is borderline and requires strong justification.
Average (Peer Norm)	150–300 pairs	Power ≥ 0.85–0.95 at medium effect with Wilcoxon + Bonferroni correction	Median range across 14 SSCI-indexed RAG-in-education papers surveyed (2022–2025). The 200-pair plan falls in the upper portion of this range.
Maximum (Tier 1 Strong)	400–600 pairs	Power > 0.95 even for small effects (d=0.3); subgroup analyses by topic/module become viable	Found in well-resourced lab studies with paid annotators. Increases annotation burden substantially. Not required for this study given the synthetic augmentation strategy.
Nyansapo Plan	200 pairs ✓	Statistically adequate for primary comparisons with Bonferroni correction	Correct and Tier 1-defensible. The outstanding gap is the annotation protocol (Cohen's Kappa ≥ 0.80), not the count. Add this protocol detail during manuscript writing (Priority A in revised assessment).

Tier 2 of the Evaluation Set: Synthetic Query Augmentation

The team plans to augment the 200 human-annotated pairs to 5,000 test queries using LLM-based query generation (the correct replacement for the originally planned CTGAN approach, which was flagged and fixed in the revised proposal). These synthetic queries serve a different function from the human-annotated pairs: they provide scale for BERTScore averaging, RAGAS metric computation, and latency benchmarking.

THRESHOLD	QUERY COUNT	RATIONALE
Minimum	500–1,000	Sufficient for mean BERTScore estimation with confidence intervals. Adequate for RAGAS evaluation. This level appears in fast-turnaround studies but is considered the lower boundary for a full journal paper.
Average (Peer Norm)	2,000–5,000	Standard range for RAG evaluation at SSCI-level venues. Enables stable latency histograms (mean, p95, std), robust BERTScore distributions, and ablation comparisons across retrieval strategies (BM25 vs Dense vs Hybrid).
Maximum (Tier 1 Strong)	10,000–20,000	Found in lab studies with GPU clusters. At 5,000 queries on Colab A100, the Nyansapo experiment is already estimated to require checkpoint saving at 500-query intervals. Exceeding 10,000 queries risks Colab session timeout and provides diminishing returns on metric stability.
Nyansapo Plan	5,000 ✓	Sits at the top of the average peer range. This is the correct number: large enough for stable metric estimation and latency profiling, small enough to complete on Colab Pro+ within the planned May 11–31 experiment window using 30-minute checkpoint saves (JSON per 500 queries).

RECOMMENDATION: The 200 human-annotated pairs + 5,000 synthetic queries structure is methodologically sound and Tier 1-defensible. The critical gap is not size — it is the annotation protocol. Adding Cohen's Kappa ($\kappa \geq 0.80$) between two independent CS lecturer annotators, with a 1-page scoring rubric, closes the only remaining dataset gap flagged in the revised assessment (Priority A).

3.3 Component C: Student Interaction Logs (HCAI User Study)

The human-centred evaluation component requires a separate analysis. This involves real students interacting with the Nyansapo system, completing NASA-TLX cognitive load scales and trust questionnaires. This component is subject to CHRPE ethics clearance and must not begin until approval is received.

THRESHOLD	PARTICIPANTS	STATISTICAL BASIS	RATIONALE
Minimum	20–30	Mann-Whitney U: power $\approx$ 0.70 for medium effect at $n=25$ per group	Accepted in pilot/proof-of-concept HCAI papers. Below this threshold, NASA-TLX comparison between RAG and Vanilla LLM conditions becomes underpowered and reviewers will object.
Average (Peer Norm)	40–80	Mann-Whitney U: power $\geq$ 0.85–0.90 for medium effect; enables subgroup	Standard range for user study components in HCAI papers at Computers & Education: AI. 40

THRESHOLD	PARTICIPANTS	STATISTICAL BASIS	RATIONALE
		by gender/department	participants is the practical minimum that Tier 1 reviewers accept without statistical power concerns.
Maximum (Tier 1 Strong)	100–150	Power > 0.95; regression modelling of trust predictors becomes viable	Achievable if CHRPE approval is received by April 30 and recruitment begins promptly in May. Provides statistical headroom for subgroup fairness analyses (gender, department) mentioned in the metrics plan.
Nyansapo Recommendation	60–80 ✓	Power ≥ 0.87 for Mann-Whitney U at medium effect; Bonferroni correction viable	This range sits solidly within the peer average and enables the gender/department fairness subgroup analysis in the metrics plan. Feasible within KNUST given the team's institutional access. Target 60 as the floor; recruit to 80 as the ceiling.

ETHICS NOTE: No student interaction log collection may begin without CHRPE approval. Michelle's ethics application (submitted March 18, approval expected April 30) is the single critical path item for this component. The recommendation of 60–80 participants is contingent on that approval being received.

4. Consolidated Dataset Requirements Summary

The table below consolidates minimum, average (peer norm), maximum, and recommended targets across all three dataset components for the Nyansapo study.

DATASET COMPONENT	MINIMUM	AVERAGE (PEER NORM)	MAXIMUM (TIER 1 STRONG)	RECOMMENDED FOR NYANSAPO
A. Retrieval Corpus (Lecture Transcripts)	50 transcripts ~500K tokens ~1,650 chunks	100 transcripts ~1M tokens ~3,300 chunks	200+ transcripts ~2M tokens ~6,600 chunks	150 transcripts ~1.5M tokens ~4,950 chunks → Above peer average; feasible via Multimedia Unit request; already planned.
B1. Gold-Standard QA Pairs (Human-Annotated)	80–100 pairs	150–300 pairs	400–600 pairs	200 pairs → Upper mid-range of peer norm; statistically adequate with Wilcoxon +

DATASET COMPONENT	MINIMUM	AVERAGE (PEER NORM)	MAXIMUM (TIER 1 STRONG)	RECOMMENDED FOR NYANSAPO
				Bonferroni; add Cohen's Kappa protocol.
B2. Synthetic Test Queries (LLM-Generated)	500–1,000 queries	2,000–5,000 queries	10,000–20,000 queries	5,000 queries → Top of average range; sufficient for RAGAS, BERTScore distributions, and latency histograms on Colab A100 with checkpoint saving.
C. User Study Participants (Student Interaction Logs)	20–30 students	40–80 students	100–150 students	60–80 students → Solidly within peer average; enables fairness subgroup analysis; contingent on CHRPE approval (expected April 30).

5. Rationale: Why These Specific Numbers?

5.1 The Retrieval Corpus: Quality Over Quantity

For a RAG system, the retrieval corpus must be large enough to contain genuine information diversity — enough distinct topics, terminology, and conceptual coverage that retrieval recall is a real challenge rather than a trivial lookup. At 150 transcripts covering Computer Science courses across multiple semesters at KNUST, the Nyansapo corpus will contain lectures on algorithms, databases, networking, software engineering, and related modules. This domain diversity is sufficient to create a genuinely challenging retrieval task.

The key insight is that FAISS retrieval quality saturates: adding transcripts beyond the point of full curriculum coverage does not improve the research contribution. 150 transcripts achieves full curriculum coverage while remaining feasible within the Multimedia Unit access request. Inflating to 300 transcripts to appear more impressive would add implementation complexity without adding scientific value — and risks losing the IP permission for FAISS indexing that Maxwell must secure.

5.2 The Human-Annotated QA Set: Statistical Power is the Binding Constraint

The 200-pair human-annotated gold standard is sized primarily by statistical power, not by an arbitrary benchmark. For the primary comparison (RAG vs. Vanilla LLM on BERTScore), a Wilcoxon signed-rank test at $n=200$ achieves power > 0.90 at medium effect sizes (Cohen's $d = 0.5$) — well above the conventional 0.80 threshold. With Bonferroni correction across three pairwise comparisons (RAG vs. Vanilla, CRAG vs. Naive RAG, Hybrid vs. Dense-only), power remains above 0.80 for medium effects.

The annotation cost is the binding constraint, not the statistical floor. Two CS lecturer annotators working from a 1-page rubric can complete 200 QA pairs in approximately 10–15 hours of annotation work spread over two weeks — feasible within the April 15–30 data preparation window. Requiring

400 pairs would double the annotator burden without meaningfully improving statistical power at the expected effect sizes.

5.3 Synthetic Query Augmentation: Scale for Metric Stability

The 5,000 synthetic queries serve a different purpose from the human-annotated pairs. They are not used to establish statistical significance — that role belongs to the 200 human-annotated pairs, where ground truth is known. Instead, the 5,000 synthetic queries enable: (a) stable BERTScore distributions with narrow confidence intervals for the Results section figures, (b) reliable latency histograms (mean, p95, standard deviation) across both online and offline inference modes, and (c) large-sample RAGAS evaluation where faithfulness, answer relevance, and context relevance scores stabilise.

Five thousand queries is the natural ceiling for the Colab Pro+ hardware constraint. At approximately 3–5 seconds per RAG query on the A100 (including retrieval, generation, and RAGAS evaluation), 5,000 queries requires approximately 4–7 hours of compute time — manageable within a single Colab Pro+ session using the planned 30-minute checkpoint saves. Exceeding 10,000 queries would require multiple sessions and reintroduces the risk of environment inconsistency across checkpoints.

5.4 User Study Sample: Balancing Power and Ethics

The recommended 60–80 participant target is grounded in two constraints: statistical power for the Mann-Whitney U test on NASA-TLX scores, and practical recruitment feasibility within KNUST. At n=60 per condition (30 per group in a between-subjects design, or 60 in a within-subjects design), Mann-Whitney U achieves power ≈ 0.87 for medium effects — adequate for the planned gender and department subgroup fairness analyses.

Importantly, the ethics clearance timeline (CHRPE approval expected April 30) creates a firm upper bound on the recruitment window. With approval on April 30 and a July manuscript submission deadline, the team has approximately 8 weeks for recruitment and data collection. At KNUST, recruiting 80 Computer Science students for an AI tutoring study is feasible within that window given the team's existing institutional relationships. Targeting 100+ participants would require an extended recruitment period that conflicts with the manuscript deadline.

6. What Tier 1 Reviewers Actually Scrutinise

Understanding the reviewer's perspective helps the team prioritise correctly. Based on editorial guidelines and reviewer report patterns at Computers & Education: AI and Frontiers in AI (2023–2025), dataset-related reviewer comments cluster into five categories:

REVIEWER CONCERN	NYANSAPO STATUS	WHAT IS NEEDED
Is the test set large enough to support significance claims?	✓ MET (200 pairs + 5,000 queries)	Report the statistical power calculation for primary comparisons. One sentence in the Methods section: 'At n=200, Wilcoxon signed-rank achieves power=0.91 (d=0.5, α=0.05, two-tailed).'
Is the annotation protocol reliable? (Inter-rater reliability)	⚠ GAP (protocol not yet described)	Add Cohen's Kappa (κ ≥ 0.80) between two CS lecturer annotators with a 1-page scoring rubric. This is Priority A in the revised assessment and must be added during manuscript writing.
Is the corpus domain-	✓ STRONG	Report word count per transcript, total token count,

REVIEWER CONCERN	NYANSAPO STATUS	WHAT IS NEEDED
appropriate?	(institutional corpus, KNUST CS)	FAISS index size, and course coverage in a corpus specification table (already drafted in Section 4.2 of the proposal).
Is synthetic data validated and not used as ground truth?	✓ MET (4-level validation plan, LLM-generated)	Report all four validation levels including k-anonymity (Level 3) for interaction logs. Clearly state that synthetic queries are used only for scale-based metric estimation, not for establishing ground truth.
Is the user study adequately powered and ethically cleared?	⚠ PENDING (CHRPE approval expected April 30)	Report power calculation for Mann-Whitney U on NASA-TLX scores. Include CHRPE approval reference number in the Ethics section. Target 60–80 participants within the 8-week post-approval recruitment window.
Is dataset availability or access described?	✓ MET (institutional access + IP permission requested)	Include the Multimedia Unit access letter reference and the FAISS IP permission notation in the Data Availability statement. Anonymised interaction logs may be deposited in a data repository (e.g., OSF) to strengthen reproducibility.

7. Final Recommendation: The Correct Sizes for Nyansapo

Based on the analysis above, the following dataset targets are recommended for the Nyansapo experiment. All three components are already planned at or near these targets, confirming that the student team's data design is sound. No substantial upsizing is required.

COMPONENT	RECOMMENDED SIZE	WHY THIS IS THE RIGHT NUMBER
Retrieval corpus (Lecture Transcripts)	150 transcripts (~1.5M tokens)	Exceeds peer average (100 transcripts). Covers full KNUST CS curriculum. Feasible via Multimedia Unit access request. FAISS index (~5,000 chunks) runs efficiently on Colab A100. Do not inflate beyond 150 without a specific scientific reason.
Human-Annotated Gold-Standard QA Pairs	200 pairs	Upper mid-range of the 150–300 peer norm. Wilcoxon power > 0.90 (d=0.5). Annotatable by two CS lecturers in ~10–15 hours total. The binding gap is the annotation protocol (Cohen's Kappa), not the count — fix the protocol, not the number.
Synthetic Test Queries (LLM-Generated)	5,000 queries	Top of the 2,000–5,000 peer average range. Large enough for stable RAGAS, BERTScore distributions, and latency histograms. Small enough to complete on Colab Pro+ with 30-minute checkpoints. Hardware-appropriate ceiling for this study.

COMPONENT	RECOMMENDED SIZE	WHY THIS IS THE RIGHT NUMBER
User Study Participants (Student Interaction Logs)	60–80 students	Within the 40–80 peer norm. Mann-Whitney U power ≈ 0.87 . Enables gender/departement subgroup analysis. Achievable in 8 weeks post-CHRPE approval. Target 60 as the floor; recruit to 80 to allow for attrition and exclusions.

THE SINGLE MOST IMPORTANT DATASET ACTION REMAINING: The dataset sizes are correct. The annotation protocol is missing. Before submission, add to Section 4.2 or Appendix: 'Two CS lecturers independently annotated each of the 200 gold-standard QA pairs using a 1-page scoring rubric defining correctness, relevance, and completeness on a 3-point scale. Inter-rater reliability was measured using Cohen's Kappa (κ = [reported value], target $\kappa \geq 0.80$). Disagreements were resolved by the supervising researcher.' This single paragraph closes the last open dataset gap flagged by the supervisor.

TEAM NYANSAPO — REVISED PROPOSAL ASSESSMENT

KNUST Research Supervision Programme · 2025–2026 · Dr. Eric Opoku Osei

REVISED SCORE	PREVIOUS SCORE	IMPROVEMENT	GRADE
129 / 145 (89%)	103 / 145 (71%)	+26 pts +18%	MERIT Approved — Proceed

✓ **SUPERVISOR DECISION: APPROVED WITH MINOR REVISIONS** — Data collection may proceed once CHRPE clearance is received (expected April 30, 2026). All Priority 1–3 actions from the first assessment have been satisfactorily addressed.

IMMEDIATE ACTION PLAN — March 18 to 31, 2026

Everything below must happen *BEFORE* the Easter exam period. These are the only research tasks for this window.

DATES	OWNER	ACTION	WHAT EXACTLY TO DO
Mar 18–19 (Wed–Thu)	ALL FIVE	Read revised proposal + this report together	Hold a 1-hour team sync (Google Meet or in person). Each person reads their assigned module sections. Confirm: Colab Pro+ subscription active, GitHub repo shared with all members, Google Drive 'Nyansapo_Manuscript_2026' folder created with section subfolders.
Mar 18–Apr 30 (ONGOING)	Michelle (HCAI Lead)	Submit KNUST CHRPE ethics application — START TODAY	CRITICAL PATH ITEM. No student interaction log collection is permitted without CHRPE approval. Submit the ethics application to chrpe@knust.edu.gh this week. Required documents: (1) this research proposal, (2) participant consent form template, (3) draft interaction log collection instrument. Expected approval: April 30, 2026. DO NOT WAIT.
Mar 18–20 (Wed–Fri)	Agyei (Lead)	Lock in Colab Pro+ and test Mistral-7B loads correctly	Subscribe to Colab Pro+ (USD 49.99/month). Open a new notebook, mount Google Drive, run: !pip install transformers bitsandbytes sentence-transformers faiss-cpu langchain. Load Mistral-7B-Instruct-v0.2 in 4-bit and verify it generates a response. Save the working notebook to GitHub. This confirms the compute environment before exam period.
Mar 18–20 (Wed–Fri)	Maxwell (Data Engineering)	Draft official letter to KNUST Multimedia Learning Unit	Write and submit a formal written request for access to 150 multimedia lecture transcripts (Computer Science courses). CC Dr. Eric Opoku Osei on the request. Also request written IP permission for FAISS indexing of the transcripts — this is required before data collection can begin after CHRPE approval.
Mar 21–25 (Mon–Fri)	Maxwell	Set up GitHub repository structure and Drive folders	Create the folder structure on Drive: 01_Introduction/, 02_RelatedWork/, 03_Methodology/, 04_Results/, 05_Discussion/, 06_Conclusion/, 07_Abstract/, 08_References/, 09_Figures/. Push the folder structure and README to GitHub (github.com/maagyei04/Nyansapo-

DATES	OWNER	ACTION	WHAT EXACTLY TO DO
			RAG-Research). All team members must confirm they can commit to the repo.
Mar 21–28 (Mon–Fri)	Maxwell (Lit)	Expand literature from 10 → 20 papers (minimum for manuscript)	For the publication-ready manuscript, 10 papers is a proposal minimum. Before the Easter break, target 20 papers. Search Scopus for: 'RAG education 2023 2024 2025', 'LLM tutoring chatbot higher education', 'AI hallucination mitigation'. Add each paper to a shared Zotero or Mendeley group library. Verify all DOIs. Aim for at least 3 new RAG application papers from 2023–2025.
Mar 26–31 (Thu–Tue)	ALL FIVE	Individual exam prep — PAUSE active experiment work	From March 26 to Easter (approx. April 17), the team should focus on examinations. The only active research tasks during this period are: (1) CHRPE application processing (Michelle monitors), (2) Multimedia Unit transcript request (Maxwell follows up), (3) Colab environment remains set up and ready. No new experiment work. Resume fully on April 15.
Mar 31 (Tuesday)	Agyei (Lead)	Quick 15-min team check-in before exam period begins	Brief status check: CHRPE submitted? YES/NO. Colab Pro+ confirmed? YES/NO. Transcript request sent? YES/NO. GitHub/Drive set up? YES/NO. These four items are the only blockers that could delay the April 15 data collection start. Resolve any NO answers before going into exam period.

Note to Dr. Osei: The team is entering an exam period after March 25. The four items that absolutely must be completed before the break are: (1) CHRPE application submitted, (2) Colab Pro+ environment verified, (3) Multimedia Unit transcript request sent, (4) GitHub/Drive folder structure ready. If all four are done, the April 15 data collection start date is protected.

REVISED SECTION-BY-SECTION SCORE COMPARISON

Every section that changed is explained below. Green shading = improvement achieved. The revised total is 129/145 (89%) — MERIT grade.

SECTION	OLD	NEW	ΔDELTA	WHAT CHANGED & WHY THE SCORE MOVED
Title & Keywords	3	5	+2	New title follows [Model]/[Task]/[Domain]/[Novelty] pattern. All 6 keywords are Scopus-indexable, including the new 'AI Hallucination Mitigation in Education' keyword.
Novelty Declaration	14	18	+4	Now falsifiable and measurable. Includes BERTScore delta hypothesis (+0.08 min), population specificity (Sub-Saharan African HE), and offline mode novelty. Minor deduction: 3rd sentence still hedges with 'hypothesised' — acceptable at proposal stage.
Research Domain Selection	5	5	0	Already perfect. Maintained.
Problem Statement	11	14	+3	Ji et al. (2023) correctly replaces the mismatched Devlin (2019) citation. Osei (2025) is now a full APA 7 entry with 'Unpublished manuscript' notation. Gap paragraph now cites Gao (2023) to confirm no prior RAG study in sub-Saharan HE. Strong improvement.
Objectives & RQs	8	10	+2	RQ1 fully revised — design assumption removed, now directly answerable by the experiment. RQ3 updated to include both online and offline modes, aligned with the new Enhancement A objective. All three RQ–Objective pairs are now tightly aligned.
Educational Theory	4	5	+1	CLT is now connected to the 300-token chunking design decision and the 7±2 working memory limit. This is the correct way to operationalise a theory in methodology. NASA-TLX added as measurement instrument in evaluation metrics — full loop from theory to measurement.
Related Work & Synthesis	10	13	+3	Gao et al. (2023) and Ji et al. (2023) correctly added as ML-relevant papers. Davis (1989) and Krejcie (1970) moved to a separate Methodological References table — excellent structural decision. Synthesis paragraph now includes the distinguishing sentence about offline inference. Minor deduction: still only 10 primary ML papers; manuscript will need 20–35 for publication.
Model Selection	9	10	+1	Mistral-7B-Instruct-v0.2 committed. Dettmers et al. (2023) cited for 4-bit NF4 quantisation. Embedding model specified (all-MiniLM-L6-v2, 384-dim). BERT-base confusion explicitly resolved in the text. Clean and complete.
Dataset Selection	7	9	+2	Corpus specification table added: word count per transcript, total token estimate, FAISS index size, 200 gold-standard human-annotated pairs added. Collection method (Whisper ASR + Multimedia Unit request) specified. Minor deduction: human annotation of 200 pairs is planned but not yet described in detail — who annotates? What inter-rater reliability protocol?
Synthetic Data Plan	3	5	+2	CTGAN correctly replaced with LLM-based query generation. All 4 validation levels ticked, including Level 3 (k-anonymity privacy audit) for student interaction logs. Excellent correction of a methodological error from the previous version.
Computing Environment	3	5	+2	Inconsistency resolved: Colab Pro+ (A100) confirmed throughout. Checkpoint saving plan specified (30-minute intervals, JSON per 500 queries). Environment feasibility confirmed for full experiment.

SECTION	O LD	N E W	ΔD ELT A	WHAT CHANGED & WHY THE SCORE MOVED
Reproducibility Checklist	6	15	+9	Outstanding improvement. 13 of 16 items now ticked with implementation detail. Previously only 6 were addressed. Specific additions: data split strategy (70/15/15 stratified), training procedure (prompt template, no fine-tuning), significance testing (Wilcoxon + Bonferroni), random seeds (5 seeds stated), library versions (full stack listed), ablation study (3 ablations defined). This is now publication-ready for reproducibility.
Ethics Clearance	4	5	+1	FAISS index IP permission correctly added. Critical sequence note added (CHRPE before data collection). k-anonymity check specified. Complete.
Results & Metrics Plan	4	5	+1	Latency metrics added for Objective 3 (mean + p95 + std deviation). NASA-TLX mental effort subscale added for CLT validation. Fairness metrics (Equal Opportunity, subgroup by gender/department) included. Comprehensive and appropriate.
Discussion & Conclusion	5	5	0	Already complete. CLT-to-findings template sentence added in Section 9.2 — this is the correct way to pre-draft the theoretical contribution before writing the full paper.
Journal Selection	3	5	+2	Target Journal 1 correctly changed to Computers & Education: AI (Elsevier). APC (~USD 1,500) with Research4Life waiver noted. Sample paper DOI provided. Turnaround (2–3 months) confirmed compatible with July deadline. Excellent execution of the Module 6 recommendation.
References	4	5	+1	Osei (2025) now fully referenced as an unpublished manuscript. Dettmers (2023), Gao (2023), Ji (2023) added. All APA 7 with DOIs. No pre-2019 papers used as primary references (Vaswani 2017 is foundational/architectural — acceptable).
TOTAL	10 3	13 9	+26	103/145 (71%) → 139/145 (96%) — Grade upgraded from PASS WITH REVISIONS to MERIT. Approved to proceed to data collection once CHRPE clearance is received.

REMAINING MINOR GAPS — For Manuscript Writing Phase (April–June 2026)

The proposal is approved. The items below are NOT blockers for data collection. They are manuscript-writing improvements to address during April–June when each section is being drafted.

P RI	SECTION	GAP	EXACT FIX (1–2 SENTENCES)
A	Dataset	Human annotation protocol not described	In Section 4.2 or Appendix, add: 'Two CS lecturers independently annotate each gold-standard QA pair. Inter-rater reliability is measured using Cohen's Kappa (target $\kappa \geq 0.80$). Disagreements are resolved by the supervising researcher.' This is a 1-paragraph addition.
B	Related Work	Only 10 primary papers (proposal minimum). Manuscript needs 20–35.	During April (before/after Easter): Maxwell adds 10–15 new papers from Scopus search: 'RAG education 2023 2024 2025'. Target: 3 RAG-in-education papers, 2 LLM tutoring papers, 2 HCAI evaluation papers. Expand synthesis to 4–5 themed paragraphs for the manuscript draft.
C	Reproducibility Item 3	Ablation study (Item 16) ticked but inter-rater reliability for 200 human-annotated pairs not mentioned	Add: 'Cohen's Kappa will be reported for human annotation agreement. Annotators will be provided with a 1-page scoring rubric defining correctness, relevance, and completeness on a 3-point scale.'
D	Discussion	Failure case analysis is ticked as optional but not yet planned	Plan 10–15 failure cases: queries where RAG retrieves wrong chunk or generates a confidently wrong answer despite retrieval. These failure cases become the 'Limitations and Future Work' section's most compelling content.
E	Novelty	Code-switching evaluation (Enhancement C from Module 2) is not included	Optional for first submission. If time permits during June user study, add 50 queries in Ghanaian English/Twi-influenced phrasing to test fairness. This is a strong future-work direction if not implemented in v1.

KEYNOTE ASSESSMENT — REVISED PROPOSAL

KEYNOTE	REVISED ASSESSMENT
Topic is Important	YES — AI hallucination reduction in Ghanaian HE is timely, policy-relevant, and practically impactful for students in high-enrollment lecture halls. SDG 4/9/10 alignment strengthens the case.
Findings are Novel	STRONG (4/5 with Enhancement A) — The offline/low-bandwidth inference mode, now included in Section 4.4, makes this the first RAG-in-education paper to address bandwidth-constrained African university deployment. Genuinely novel.
Abstract is Clear	NOT YET WRITTEN — Draft using the CLT framing template: Problem (1 sentence) → Gap (1 sentence) → Method (2 sentences: RAG + offline mode) → Key results (2 sentences: BERTScore delta + latency) → Implication (1 sentence). Write LAST.
Introduction is Clear	STRUCTURE PLANNED — Para 1 should state the contribution explicitly in the final sentence. Use the novelty statement from Section 0 as the starting point for para 1's closing contribution claim.
Prior Work is Comprehensive	PARTIAL (proposal stage) — 10 primary papers is sufficient for proposal approval. The manuscript needs 20–35. Maxwell begins expansion in April. Target: 3 RAG papers, 2 LLM-tutoring papers, 2 HCAI evaluation papers, remaining Ghana/Africa EdTech context.
Research Aims are Clear	YES — 3 objectives with fully revised, answerable RQs. RQ3 now includes both online and offline modes.
Data is Appropriate	YES — Institutional corpus is the right choice for this research question. The 200 human-annotated pairs add Tier 1 credibility. Annotation protocol (Cohen's Kappa) to be added during manuscript writing.
Analysis is Appropriate	YES — BERTScore + ROUGE-L + Wilcoxon + Bonferroni + NASA-TLX + Mann-Whitney U is a comprehensive and well-justified analysis plan. The latency metrics (mean + p95 + std) directly answer RQ3.
Implications are Supported	YES — Theoretical (CLT operationalised via NASA-TLX) and practical (React Native deployment, 3G latency testing, Ollama offline mode) implications are both present and connected to the experiment design.
References are Complete	YES — All 14 references have APA 7 formatting with DOIs. Osei (2025) now fully cited. Dettmers (2023) and Gao (2023) correctly added. No missing entries.
Manuscript is Well Structured	YES — The section structure is publication-ready. Follow the merge order: Introduction → Related Work → Methodology → Results → Discussion → Conclusion → Abstract (last) → References.
Manuscript is Well Written	GOOD at proposal stage. The main risk is terminology inconsistency across group-written sections. Run the Claude terminology audit (Module 5C) after Draft 1 compilation.
Presentation is Clear	YES — All planned figures (BERTScore chart, latency histogram, attention heatmap, user trust table, ablation table, t-SNE) are appropriate and sufficient for the target journal.
Tables are Adequate	PLANNED — Metric comparison table + ablation table + corpus specification table (already drafted in Section 4.2). Add system architecture diagram (React Native → FastAPI → FAISS → Mistral-7B flow).
Figures are Adequate	PLANNED — Add one system architecture figure before submitting to C&E:AI. HCAI reviewers expect to see how a human interacts with the AI system, not just the ML pipeline.

A Final Word to Team Nyansapo

From the first proposal to this revision, you have done something that many research teams struggle to do: you took detailed critical feedback, understood why each change mattered, and implemented every

one of them correctly. The jump from 71% to 89% in a single revision cycle is not a small thing — it reflects genuine intellectual engagement with the research process. The novelty is now real. The reproducibility is now publication-grade. The experiment is feasible, the ethics plan is responsible, and the journal choice is right. Between now and the end of March, your only job is to submit the CHRPE application, confirm the Colab environment, and send the transcript request. Then go and do well in your exams — Nyansapo will be waiting on the other side of Easter, ready to be built. The research community needs this work done, and you are the right team to do it.

Draft 0_ Experiment (SEE FOUR REPORTS BELOW)

SUPERVISOR DASHBOARD

Dear Team Nyansapo,

I have reviewed the Draft 0 Methods and Results sections against Q1 review standards. The direction of the work remains strong and the offline-deployment framing is genuinely novel, but the current Results section has two issues that must be fixed before this goes anywhere near a journal, and I want to flag them clearly rather than let them travel further into the manuscript.

First, the RAG evaluation (BERTScore, ROUGE-L, the 5-fold CV table, and the topic-fairness table) scores Nyansapo CRAG against its own generated answers rather than against the independent gold-standard QA pairs. That guarantees a perfect score and isn't a valid comparison — your own note at the end of the Results section already flags that this needs to be replaced with human-verified references, so let's do that now rather than at camera-ready stage. Second, the baseline classifier numbers in Table 4 (the seed-42 example) are not statistically consistent with the aggregate mean and standard deviation reported in Table 5 across all 20 seeds — the gap is large enough that the two tables cannot both be correct as written, and I need this traced back to the actual run logs before we go further.

Please pull the raw run outputs for both issues and meet as team to deliberate on the report.

REPORT 1: EXPERIMENTAL METHOD AND RESULTS

STUDENT/TEAM: Group 1 — Nyansapo

MODEL: Nyansapo-QIC-WCE intent classifier (weighted cross-entropy feedforward net) feeding a hybrid BM25+dense CRAG pipeline (Phi-2 online / Phi-4-mini offline)

CLAIM TYPE: Mismatch — the title and RQs bound the study to a Ghanaian university deployment context (locally-scoped, type ii), but the WCE and CRAG performance claims are argued and evaluated as if they were generalizable methodological contributions (type i), with no external dataset, no sensitivity analysis, and no explicit limitations statement scoping generalizability.

VERDICT: Reject / Fundamental Flaw (Do Not Submit) in current form

GENERALIZATION GATE: MISMATCH — locally-scoped deployment framing paired with universal-sounding "the engineered model achieves..." claims, validated only on a 145-example in-house dataset with no held-out site, no perturbation testing, and no honest limitations statement.

WORST FLAG: The RAG evaluation (BERTScore, ROUGE-L, 5-fold CV, and the fairness-by-topic table) uses "Nyansapo CRAG answers as the reference standard" — meaning the proposed system is scored against its own output. A perfect score of 1.0000 with zero variance across 100 queries and 5 folds is not a result, it is a tautology, and the team's own footnote in section 3.5 admits this ("replacing CRAG self-reference scores with human-verified reference scores" in the camera-ready version) — confirming they know the current numbers are not real evaluation results.

DETAILED ANALYSIS

1. Data Integrity & Preprocessing Scrutiny — Fatal

Flag: Self-referential ground truth (circular evaluation)

Evidence: "Using Nyansapo CRAG answers as the reference standard, the system achieves

a perfect BERTScore of 1.000" and "Nyansapo CRAG achieves ROUGE-L = 1.000 across all eight topic areas."

Q1 Reviewer Critique: Scoring a system against its own outputs guarantees a perfect score by construction. This is not a comparative evaluation, it is measuring the identity function. Any Q1 reviewer with NLP evaluation experience will flag this as either a fundamental misunderstanding of the metric or an attempt to manufacture a superiority result. The fact that CRAG's Faithfulness score (0.591) is simultaneously reported as lower than Naive RAG's (0.753) while BERTScore/ROUGE-L show CRAG "winning" by construction is itself an internal contradiction — the two metric families cannot both be honest if one is rigged toward the proposed system.

Strict Fix Required: Re-run BERTScore, ROUGE-L, and the fairness-by-topic and 5-fold-CV analyses against the 200 independent human-annotated gold-standard QA pairs described in the Methods section, not against CRAG's own generations. Every table currently built on CRAG-as-reference (Tables 7, 8, 9, 12, and the consolidated Table 14) must be discarded and regenerated. Until this is done, RQ2 has no valid result.

Flag: Methods/Results split-count inconsistency

Evidence: Methods Table 2 states the stratified test fold is $n=22$ (Definition=13, Explanation=3, Identification=0, Procedural=4, Out-of-scope=2). Results Table 4 reports the seed-42 test fold as $n=29$ (Definition=16, Explanation=5, Identification=1, Procedural=5, Out-of-scope=2).

Q1 Reviewer Critique: These are not rounding differences — the total sample size differs by 7, and the Identification class goes from 0 examples (per the stated 70/15/15 stratified protocol) to 1 example in the reported classification report. Either the stratified split actually used in the experiment was never applied as documented, or the numbers in one of the two sections were constructed after the fact rather than measured. A reviewer who cross-checks Methods against Results — which is standard practice — will catch this immediately and it reads as fabricated internal consistency.

Strict Fix Required: Regenerate the actual split from the code, report the true per-class counts once, and make the Methods and Results sections reference the identical split. If the split changed for a legitimate reason (e.g., a later re-shuffle), state that explicitly and justify it.

4. Evaluation Metrics & Results Presentation — Fatal

Flag: Statistically impossible per-seed vs. aggregate mismatch

Evidence: Table 4 (seed 42) reports a baseline Macro F1 of 0.67 with per-class F1 values of 0.86, 0.75, 0.00, 0.75, 1.00. Table 5 reports the mean baseline Macro F1 across all 20 seeds (including seed 42) as 0.1802 ± 0.0459 .

Q1 Reviewer Critique: If seed 42 genuinely scored 0.67 and the reported standard deviation across 20 seeds is 0.0459, seed 42 sits roughly 10 standard deviations above the claimed mean. That is not a plausible single sample from that distribution — it indicates the aggregate statistics in Table 5 were not actually computed from the per-seed results in Table 4, or Table 4 is illustrative/invented rather than a genuine seed-42 run. Either way, the numbers cannot both be true. Reviewers who spot-check summary statistics against example tables (again, standard practice) will treat this as evidence of fabricated results, not a typo.

Strict Fix Required: Recompute both tables from the same underlying run logs and confirm they are mutually consistent before submission. If Table 4 is meant to illustrate "a representative failure case" rather than seed 42 specifically, label it that way explicitly and do not claim it is seed 42.

Flag: Narrative text contradicts its own table

Evidence: The text claims the baseline "achieves $F1 = 1.00$ for the dominant Definition class," but Table 4 lists Definition $F1 = 0.86$ for that same run.

Q1 Reviewer Critique: A prose claim that doesn't match the adjacent table is a red flag for either careless copy-paste from a different run or numbers that were narrated before they were finalized. Both are integrity concerns at Q1 level.

Strict Fix Required: Align narrative claims to the exact table values in the final draft, and add a verification pass comparing every quoted number in prose to its source table.

Flag: "Too perfect" curves / zero-variance results

Evidence: Table 9 reports CRAG ROUGE-L = 1.000 with SD = 0.000 across all five folds and all five seeds.

Q1 Reviewer Critique: Zero variance across five independent folds on a generative NLP task is not a strong result, it is a red flag for measurement circularity (consistent with the self-reference issue above). No legitimate LLM-based generation pipeline produces literally identical scores across folds unless it is being compared to itself.

Strict Fix Required: Same fix as the self-reference flag above — once evaluation is against the independent gold-standard set, genuine variance should reappear, and if it doesn't, that itself needs investigation rather than being presented as a strength.

Flag: RAGAS Faithfulness explanation is inconsistent with the stated methodology

Evidence: Results section attributes CRAG's lower Faithfulness score (0.591 vs. Naive RAG's 0.753) to "the keyword-overlap implementation of the Faithfulness scorer used in this study." But Methods section 2.5.2 describes RAGAS Faithfulness as computed via the actual `ragas` library (an LLM-as-judge metric per Es et al., 2023), and separately describes the CRAG evaluator's internal retrieval-quality gate (not the Faithfulness metric) as keyword-overlap based.

Q1 Reviewer Critique: This is a post-hoc rationalization that conflates two different components (the CRAG evaluator's internal gate vs. the RAGAS Faithfulness scoring function) to explain away an inconvenient result — the proposed anti-hallucination system scoring worse than the naive baseline on the paper's own primary anti-hallucination metric. If RAGAS is genuinely LLM-judged as stated in Methods, this explanation is simply incorrect, and the real result (CRAG underperforming Naive RAG on Faithfulness) stands unexplained and undermines the central thesis of the paper.

Strict Fix Required: Either correct the explanation to accurately describe why CRAG underperforms on Faithfulness (a legitimate and analyzable phenomenon — e.g., query rewriting shifting retrieved vocabulary), or accept and report the result as a genuine limitation rather than reframing it as a scorer artifact.

5. Reproducibility & Environment Transparency — Major

Flag: Field validation dataset promised in Methods, absent from Results

Evidence: Methods section 2.3.2 states field validation Macro F1 and out-of-scope F1 "are reported alongside the stratified test fold results." No such data appears anywhere in the Results section.

Q1 Reviewer Critique: A promised independent validation set that silently disappears between Methods and Results, particularly one meant to demonstrate real-world generalization beyond the training distribution, will draw a direct reviewer question about whether the CHRPE-approved field study (June 11–20) actually occurred and whether results were dropped because they were unfavorable.

Strict Fix Required: Either report the field validation results as planned, or explicitly state in

Limitations that CHRPE-gated data collection had not concluded at manuscript submission and give a timeline.

Gate Check — Field-to-Public Generalization

Step A classification: The title and RQ framing bound the work to "a Bandwidth-Constrained Ghanaian University Environment" (locally-scoped, type ii), while RQ1 and the ablation study argue the weighted cross-entropy engineering operation as a generalizable contribution ("confirming that the weighted cross-entropy loss function... is the sole source of the performance improvement") without qualifying that this is demonstrated only on a single 145-example, single-institution dataset.

Step B: Under the locally-scoped reading, the required substitute rigor is only partially met.

Present: multiple seeds (20) and an ablation isolating the mechanism. Missing: any sensitivity/perturbation analysis showing the classifier doesn't collapse under small input shifts (e.g., paraphrased or code-switched queries — the team's own supervisor notes flag Ghanaian-English/Twi-influenced phrasing as unaddressed future work), and no explicit limitations sentence scoping the claim to KNUST/this classifier only. Under the universal reading, no external or held-out institution's data is used at all, which would cap the paper outright.

Recommendation: State explicitly in Limitations that generalizability of the WCE weighting scheme and the CRAG pipeline beyond this single institution, single 145-example training set, and single corpus of 10,865 lecture-slide chunks has not been established, and add at minimum a perturbation test (paraphrased/code-switched query variants) before claiming the engineering contribution is robust.

Dimensions checked: 1,2,3,4,5 — clean: 2 (Baseline & Comparative Fairness — the four RAG systems and the QIC ablation conditions do appear to use matched preprocessing and prompts), 3 (Methodological & Architectural Illusions — the WCE loss modification is a real, specified mathematical change, not buzzword-only, and the CRAG/RRF/BGE-M3 components are correctly attributed to their source papers, not misrepresented as novel).

HANDOFF TO PROMPT 2

CLAIM TYPE: Mismatch (locally-scoped title/RQs, universal-sounding WCE and CRAG claims)
FLAG REPORT:

1. Data Integrity — Self-referential ground truth (CRAG scored against its own outputs) — Severity: Fatal — Invalidates every RAG-comparison table (7, 8, 9, 12, 14).
2. Evaluation Metrics — Statistically impossible per-seed vs. aggregate F1 mismatch (seed-42 table vs. 20-seed mean) — Severity: Fatal — Suggests the two tables were not computed from the same run data.
3. Data Integrity — Methods-vs-Results test-split count mismatch (n=22 vs. n=29, Identification class 0 vs. 1) — Severity: Major — Indicates the documented split protocol was not actually followed.
4. Evaluation Metrics — Narrative text contradicts its own results table (Definition F1 claimed as 1.00, table shows 0.86) — Severity: Major.
5. Evaluation Metrics — Zero-variance ("too perfect") CRAG scores across 5 folds/5 seeds — Severity: Major — Same root cause as flag 1.
6. Reproducibility — RAGAS Faithfulness result contradicted by post-hoc explanation inconsistent with stated methodology — Severity: Major — Undermines the paper's central anti-hallucination claim.
7. Reproducibility — Promised field-validation dataset absent from Results — Severity: Minor — Needs explicit acknowledgment or completion before submission.

8. Gate — Under-defended local claim, missing perturbation analysis and explicit limitations scoping statement — Severity: Major.

REPORT 2: HOW TO FIX SUGGESTIONS

MUST-FIX MODULES (Fatal and Major flags, in Flag Report order)

Note before I start: none of the actual notebook code or column names for Group 1 were included in what I've been given — only the Methods PDF, the audit PDF, the Prompt 1 lens, and the Results docx with narrative tables. Per the guardrail in Rule 3, I am not going to invent variable names that look like they came from your repo. Below I use clearly-marked generic placeholders (your_reference_answers, your_seed_log, etc.) and I am asking that the actual scoring script and the seed-results CSV/dataframe be shared at the next meeting so the code samples can be made truly copy-pasteable.

FLAG 1 — Self-referential ground truth in RAG evaluation (Severity: Fatal)

1. The Trap — What went wrong and why it's dangerous

I know the intent was to show Nyansapo CRAG performing best across BERTScore, ROUGE-L, the 5-fold CV, and the topic-fairness table. But scoring CRAG's answers against CRAG's own answers as the "reference" guarantees a perfect 1.000 score every time, on every query, in every fold — not because the system is good, but because you built a tautology. A Q1 reviewer sees a BERTScore of exactly 1.0000 with SD = 0.0000 across 100 queries and immediately knows the comparison is circular, because no real generative system scores a perfect, zero-variance match to an independent reference. This isn't a formatting issue — reviewers treat manufactured perfect scores as a data-integrity violation, and it puts every other result in the paper under suspicion, including the ones that are actually sound (like the WCE classifier improvement).

2. The Insider Fix — The Q1 Standard

Every legitimate RAG evaluation paper scores system outputs against an independent, human-authored (or at minimum human-verified) reference answer — never against the system's own generation. You already have exactly this asset planned: the 200 human-annotated gold-standard QA pairs described in your Methods section 2.3.3, with a target Cohen's Kappa ≥ 0.80 between two CS lecturers. That is your reference standard. The 100 LLM-generated queries used for BERTScore/ROUGE-L in the Results section need to be scored against gold answers from that set (or a subset of it), not against CRAG's own output.

3. Step-by-step tutorial

Because I don't have your actual scoring script, here is the structural fix in generic terms — please swap in your real object names when you send the notebook:

```
your_gold_reference_answers = load_gold_standard_qa_pairs() # the 200 human-annotated pairs
from Methods 2.3.3, NOT CRAG's own output
your_system_outputs = {
    "crag": generate_answers(nyansapo_crag_pipeline, your_test_queries),
    "naive_rag": generate_answers(naive_rag_pipeline, your_test_queries),
    "bm25_only": generate_answers(bm25_only_pipeline, your_test_queries),
```

```
"vanilla_llm": generate_answers(vanilla_llm_pipeline, your_test_queries),
}
```

The critical fix: reference is the human-annotated gold answer, never a system's own output

```
for system_name, outputs in your_system_outputs.items():
    bertscore_result = bert_score.score(
        cands=outputs,
        refs=your_gold_reference_answers, # <- this must be the gold set, not crag's own answers
        lang="en"
    )
    rouge_result = rouge_scorer.score_batch(outputs, your_gold_reference_answers)
```

Every table currently built on CRAG-as-reference (Tables 7, 8, 9, 12, 14) needs to be regenerated with this reference source. If you only have 100 or 200 gold pairs and need 5,000 queries for the RAGAS/latency work, that's fine — the gold set anchors BERTScore/ROUGE-L, and the larger LLM-generated set can still support RAGAS Faithfulness (which doesn't require a gold answer, it checks groundedness against retrieved context).

4. Manuscript rewrite

Replace the current 3.2.2 opening with something like: "BERTScore F1 was computed against the 200 human-annotated gold-standard reference answers described in Section 2.3.3, independently of any system's own output, to avoid circularity in the comparison. Nyansapo CRAG achieved a mean BERTScore F1 of [X], significantly higher than the Naive RAG baseline ([Y]), BM25-only ([Z]), and Vanilla LLM ([W]) conditions (Wilcoxon signed-rank, Bonferroni-corrected)." Leave the bracketed values as placeholders until you've re-run the scoring — do not carry forward the 1.0000 figures.

5. Integrity check

Before you re-run this: if you re-score CRAG against the gold answers and it no longer wins by a clean, statistically significant margin over Naive RAG, are you prepared to report that honestly as the actual finding, even if it's a less dramatic story than "1.000 across the board"?

FLAG 2 — Statistically impossible per-seed vs. aggregate F1 mismatch (Severity: Fatal)

1. The Trap

Table 4 (labeled "seed 42") shows a Macro F1 of 0.67 for the baseline classifier. Table 5 reports the mean baseline Macro F1 across all 20 seeds — including seed 42 — as 0.1802 with SD 0.0459. A value of 0.67 is roughly ten standard deviations from that mean. That is not a coincidence a reviewer will let slide; it means the two tables were not computed from the same underlying run, and once one number in a results section is shown to be inconsistent with another, reviewers stop trusting the rest of the paper's numbers by default.

2. The Insider Fix — The Q1 Standard

Every number in a results section needs to trace back to one canonical run log, and that log needs to be the single source for both the illustrative single-seed table and the aggregate statistics. Q1-level reproducibility means anyone on your team (or a reviewer, in principle) could regenerate Table 4 and Table 5 from the same CSV and get the same numbers.

3. Step-by-step tutorial

```
your_seed_log = pd.read_csv("baseline_qic_20seed_results.csv") # replace with your actual
results file
```

Sanity check BEFORE writing any table into the manuscript:

```
seed_42_macro_f1 = your_seed_log.loc[your_seed_log["seed"] == 42, "macro_f1"].values[0]
aggregate_mean = your_seed_log["macro_f1"].mean()
aggregate_sd = your_seed_log["macro_f1"].std()

z_score = (seed_42_macro_f1 - aggregate_mean) / aggregate_sd
print(f"Seed 42 Macro F1: {seed_42_macro_f1}, aggregate mean: {aggregate_mean}, z-score: {z_score}")
assert abs(z_score) < 3, "Seed 42 result is inconsistent with the aggregate — do not write this table until resolved."
```

Run this assertion for every seed before any table goes into a draft. If seed 42's true logged value is actually close to 0.18 (not 0.67), then Table 4 was built from a different run, a different fold, or hand-edited numbers — track down which, and rebuild it from the actual seed-42 output.

4. Manuscript rewrite

Once corrected, the framing in 3.1.1 should read: "Table 4 presents the per-class classification report for a representative seeded run of the baseline Nyansapo-QIC (seed [actual seed used], Macro F1 = [value consistent with the aggregate]), illustrating the class-collapse pattern that is confirmed at the aggregate level (mean Macro F1 = 0.1802 ± 0.0459 across 20 seeds, Table 5)." Only pick an illustrative seed once you've confirmed its value is plausible under the aggregate distribution.

5. Integrity check

Ask yourself: did I pick seed 42 for Table 4 because it was the actual first seed in the list, or because it happened to produce numbers that told a cleaner story? If it's the latter, that's cherry-picking a result, and the fix isn't just numerical consistency — it's choosing which example to show based on representativeness, not favorability.

FLAG 3 — Methods-vs-Results test-split count mismatch (Severity: Major)

1. The Trap

Your Methods section (Table 2) says the stratified 70/15/15 split gives a test fold of $n=22$, with the Identification class getting zero test examples. Your Results section (Table 4) reports a test fold of $n=29$ with Identification getting 1 example. A reviewer who cross-references Methods against Results — which is routine — will conclude that the split you described was not the split you actually ran, which raises the question of what else in Methods wasn't followed as written.

2. The Insider Fix

The split used in the code must be the split described in the manuscript, full stop. If the split logic evolved (e.g., you switched from a fixed 70/15/15 to a stratified-with-minimum-per-class rule so Identification wouldn't vanish entirely from testing), that's a legitimate and even sensible change — but it has to be described accurately, not silently left inconsistent between sections.

3. Step-by-step tutorial

```
from sklearn.model_selection import train_test_split
```

Confirm exactly what split logic actually produced your test fold:

```

your_train_df, your_temp_df = train_test_split(
your_intent_dataset, test_size=0.30, stratify=your_intent_dataset["intent_label"],
random_state=42
)
your_val_df, your_test_df = train_test_split(
your_temp_df, test_size=0.50, stratify=your_temp_df["intent_label"], random_state=42
)
print(your_test_df["intent_label"].value_counts())

```

Compare this printed output directly against Table 2 AND Table 4 before either is finalized.

If the printed counts match neither table, that tells you the manuscript tables were built from an earlier or different run than the one currently in your repo.

4. Manuscript rewrite

Update Table 2 and Table 4 to report the same n and the same per-class counts, and add one sentence to 2.3.1: "Note: because the Identification class contains only 5 total examples, a strict 70/15/15 split would leave it entirely unrepresented in testing; the split was therefore adjusted to guarantee at least 1 test example per class, yielding the counts in Table 2." Only include this sentence if it's actually true of what you ran — otherwise, just fix the split to genuinely match what's reported.

5. Integrity check

Can you point to the single script or notebook cell that produced Table 2's numbers, and confirm it's the same cell that produced Table 4's numbers? If not, that's the first thing to resolve before anything else in this flag.

FLAG 4 — Narrative text contradicts its own results table (Severity: Major)

1. The Trap

The Results text states the baseline "achieves F1 = 1.00 for the dominant Definition class," but Table 4 shows Definition F1 = 0.86. This is a small-looking error, but it's exactly the kind of detail a careful reviewer checks first, because it's fast to verify and immediately suggests the prose was written before the final numbers were locked in.

2. The Insider Fix

Every quoted number in prose should be generated programmatically from the same table it's describing, or at minimum copy-pasted directly from the final table right before submission — not written from memory of an earlier run.

3. Step-by-step tutorial

Generate the sentence directly from the table so it can never drift from it

```

definition_f1 = your_classification_report_df.loc["Definition (Class 0)", "F1-Score"]
print(f"The baseline achieves F1 = {definition_f1:.2f} for the dominant Definition class.")

```

Use this pattern (pull the exact number out of the dataframe into an f-string) for every specific metric quoted in prose, rather than typing the number by hand.

4. Manuscript rewrite

"The baseline exhibits severe class-collapse behaviour: it achieves F1 = 0.86 for the dominant Definition class but F1 = 0.00 for the Identification class and near-zero

performance on other minority classes" — matching Table 4 exactly once Table 4's numbers are themselves confirmed consistent (see Flag 2).

5. Integrity check

Before your next full read-through, will you do a line-by-line pass comparing every number in the prose to its source table, rather than trusting your memory of what the results "were"?

FLAG 5 — Zero-variance, "too perfect" CRAG scores across folds (Severity: Major)

1. The Trap

Table 9 shows CRAG achieving ROUGE-L = 1.000 with SD = 0.000 across all 5 folds and all 5 seeds. This is the same root cause as Flag 1 (self-reference), but it deserves its own fix because zero-variance results are a red flag independent of the circularity — even a genuinely excellent system shows some variance across folds and seeds if the reference is external and the generator has any stochasticity (temperature > 0, retrieval ties, etc.).

2. The Insider Fix

Once you fix the reference-standard problem in Flag 1, re-run the 5-fold CV against the gold-standard answers. Expect and report real variance — variance is not a weakness in this context, it's evidence the evaluation is measuring something real.

3. Step-by-step tutorial

```
fold_scores = []
for fold_seed in [42, 123, 456, 789, 1024]:
    your_fold_test_set = get_fold(your_gold_reference_answers, seed=fold_seed)
    score = rouge_scorer.score_batch(
        generate_answers(nyansapo_crag_pipeline, your_fold_test_set["queries"]),
        your_fold_test_set["gold_answers"] # external reference, not CRAG's own output
    )
    fold_scores.append(score)

print(f"CRAG ROUGE-L across folds: mean={np.mean(fold_scores):.3f},
SD={np.std(fold_scores):.3f}")
```

A non-zero SD here is expected and should be reported as-is, not treated as a problem to fix.

4. Manuscript rewrite

"Table 9 presents the 5-fold cross-validation ROUGE-L scores against the human-annotated gold-standard answers. Nyansapo CRAG achieves a mean ROUGE-L of [X] (SD = [Y]), outperforming Naive RAG ([Z], SD=[W]) and Vanilla LLM ([...]) across all five folds, with the observed variance reflecting genuine query-level difficulty differences rather than measurement artifact."

5. Integrity check

If the corrected numbers show CRAG winning by a smaller, noisier margin instead of a clean sweep — is that still a paper you're proud to submit? (It should be — a real, modest, honestly-measured improvement is far more publishable than a fabricated perfect one.)

FLAG 6 — RAGAS Faithfulness result contradicted by an inconsistent post-hoc explanation (Severity: Major)

1. The Trap

CRAG scores lower than Naive RAG on RAGAS Faithfulness (0.591 vs 0.753) — which directly undercuts your central hallucination-reduction claim — and the Results section

explains this away by attributing it to "the keyword-overlap implementation of the Faithfulness scorer," conflating your CRAG evaluator's internal keyword-overlap gate (Methods 2.4) with the actual RAGAS Faithfulness metric, which your own Methods section 2.5.2 says is computed via the `ragas` library's LLM-as-judge implementation. A reviewer who reads both sections back to back will see that the explanation doesn't hold up, which reads as motivated reasoning to protect the paper's thesis rather than an honest account of a genuine finding.

2. The Insider Fix

An inconvenient result that contradicts your hypothesis is not something to explain away with a mismatched technical justification — it's something to investigate and report honestly, ideally with a real diagnostic (e.g., manually inspecting 10–15 CRAG answers where Faithfulness was low, to see whether the query-rewriting step is genuinely pulling in less-relevant context).

3. Step-by-step tutorial

```
low_faithfulness_cases = your_ragas_results_df[
    (your_ragas_results_df["system"] == "crag") & (your_ragas_results_df["faithfulness"] < 0.5)
]
for _, row in low_faithfulness_cases.head(10).iterrows():
    print(f"Original query: {row['original_query']}")
    print(f"Rewritten query (if triggered): {row['rewritten_query']}")
    print(f"Retrieved chunks: {row['retrieved_chunks']}")
    print(f"Generated answer: {row['generated_answer']}\n")
```

Manually read these to determine the REAL cause: does query rewriting change retrieved vocabulary in a way

that genuinely reduces groundedness, or is something else going on?

4. Manuscript rewrite

"Contrary to expectation, Nyansapo CRAG achieved a lower mean RAGAS Faithfulness score (0.591) than the Naive RAG baseline (0.753). Manual inspection of [N] low-Faithfulness CRAG cases suggests [actual diagnosed cause, e.g., "the query-rewriting step occasionally retrieves semantically related but lexically distinct chunks, which RAGAS's claim-decomposition step penalizes despite the answer remaining substantively correct"]. This is reported as a genuine limitation of the current CRAG threshold tuning ($\theta = 0.45$) rather than a scorer artifact, and is discussed further in Section 4."

5. Integrity check

Are you willing to let this result stand as a real, reportable limitation of your system, even though it complicates the "CRAG reduces hallucination" headline — rather than reaching for an explanation that makes the inconvenient number disappear?

FLAG 8 — Gate: under-defended locally-scoped claim (Severity: Major)

Per Rule 4: your claim type from Prompt 1 was Mismatch/locally-scoped-under-defended, not a Gate PASS, so this module is warranted — it is not manufacturing a demand for public data you don't need; it's asking for the internal rigor a locally-scoped claim still owes the reader.

1. The Trap

Your title and RQs correctly scope the work to a Ghanaian university deployment context — that's legitimate and you don't need an external public benchmark to publish a locally-

scoped study. But the way RQ1 and the ablation are written ("confirming that the weighted cross-entropy loss function... is the sole source of the performance improvement") reads as a general engineering claim about WCE, not just a KNUST-specific finding, and you haven't yet backed that broader-sounding claim with the internal rigor it needs: no sensitivity/perturbation testing, and no explicit sentence limiting the claim's scope.

2. The Insider Fix — The Q1 Standard

For a legitimately locally-scoped study, Q1 reviewers accept the absence of external data if you (a) show the result survives small input perturbations, and (b) explicitly say, in your own words, where the claim's boundaries are. This is cheaper than chasing an external dataset and it's the methodologically honest path here.

3. Step-by-step tutorial

```
import random
```

Sensitivity/perturbation test: does the WCE classifier collapse under small, realistic query variation?

```
your_perturbed_queries = []
for query in your_test_queries:
    your_perturbed_queries.append(paraphrase_or_lightly_reword(query)) # e.g., synonym swap,
    reordering
    your_perturbed_queries.append(add_ghanaian_english_phrasing(query)) # e.g., Twi-influenced
    phrasing,
    # flagged as future work by your
    # supervisor's audit — do it now instead
perturbed_macro_f1 = evaluate(nyansapo_qic_wce_model, your_perturbed_queries,
    your_true_labels)
print(f"Macro F1 on perturbed queries: {perturbed_macro_f1} vs. original: {original_macro_f1}")
```

A modest, expected drop is fine and expected. A collapse back toward baseline (~0.18) means the WCE

improvement was fit to the exact training phrasing, not the underlying task.

4. Manuscript rewrite

Add to Limitations: "This study demonstrates the weighted cross-entropy engineering operation within a single institution's query distribution (KNUST Computer Science). A perturbation analysis on paraphrased and Ghanaian-English-influenced query variants ([N] samples) showed [result]. Generalizability of the specific class weights and the underlying WCE formulation to other institutions, disciplines, or query distributions has not been established and is a direction for future work." This single addition converts a mismatch into an honestly-scoped, defensible local claim.

5. Integrity check

If you ran the perturbation test right now, do you expect Macro F1 to hold up reasonably well, or do you suspect it might collapse? If you suspect the latter, that's worth knowing before submission, not after review.

POLISH MODULE (Minor flags, bundled)

Flag 7 — Field validation dataset promised in Methods but absent from Results.

Fix: Either report the field validation Macro F1 and out-of-scope F1 from the June 11–20 user

trust study now that CHRPE approval has presumably progressed, or add one sentence to Limitations stating the field validation data collection had not concluded at the time of this draft and giving an expected completion date. Don't let a promised dataset just silently vanish between sections — a single clarifying sentence closes this gap.

FINAL SUPERVISOR SUMMARY AND NEXT STEPS

Must Fix Immediately (Fatal/Major) — do not proceed to journal formatting until these are resolved and re-reviewed:

- Flag 1: Replace CRAG-as-its-own-reference scoring with the human-annotated gold-standard answers for BERTScore, ROUGE-L, 5-fold CV, and topic-fairness tables.
- Flag 2: Trace Table 4 and Table 5 back to one consistent run log; resolve the seed-42-vs-aggregate statistical impossibility.
- Flag 3: Reconcile the Methods (n=22) and Results (n=29) test-split counts to the actual split used.
- Flag 4: Correct the Definition-class F1 narrative to match the (corrected) table.
- Flag 5: Re-run 5-fold CV against the external reference and report genuine variance.
- Flag 6: Replace the mismatched RAGAS Faithfulness explanation with a genuine diagnostic of the low-Faithfulness cases.
- Flag 8: Add a perturbation analysis and an explicit limitations sentence scoping the WCE/CRAG claims to this institution's data.

Should Fix for Polish (Minor) — address in final proofing:

- Flag 7: Report or explicitly explain the absence of the field validation results.

Suggested checkpoint: bring the corrected run logs, the regenerated Tables 4–14, and the perturbation-analysis results to our next meeting so we can re-review before any journal formatting begins.

WHAT WOULD MAKE THIS A GLOBAL, GENERALIZABLE Q1 CONTRIBUTION RATHER THAN A LOCAL CLAIM ONLY

- Run the same WCE-vs-standard-cross-entropy ablation on at least one publicly available intent-classification or educational-query dataset (even a small one) to show the weighting scheme's benefit isn't an artifact of your specific 145-example set.
- Add the perturbation/sensitivity test above (paraphrased and Ghanaian-English-influenced queries) as standard practice, not just a one-off fix — this is exactly the kind of robustness check that turns "it worked on our data" into "it's robust to realistic variation."
- Once the KNUST IDL Whisper-transcribed corpus replaces the slide-based development corpus (already planned per your Section 2.3.3 note), report both corpora's results side by side — that internal replication across two corpora from the same institution is a meaningful step toward generalizability even before you reach another institution.
- If a second institution, department, or even a different KNUST cohort/semester becomes available later, re-running the CRAG pipeline there — unchanged, no re-tuning — would be the strongest possible generalization evidence and is worth planning for as a fast-follow paper.
- Frame the offline/low-bandwidth deployment mode explicitly as the generalizable contribution (it's genuinely novel and portable to any bandwidth-constrained setting, not

just KNUST), while keeping the WCE classifier and CRAG-tuning specifics honestly scoped to this dataset until validated elsewhere.

This is a strong foundation — the offline-deployment idea and the WCE engineering logic are both real, well-motivated contributions, and every flag above is fixable without touching what's actually good about this project. Catching the self-reference issue and the table inconsistencies now, before submission, is exactly what turns this from a paper that gets desk-rejected into one that gets seriously reviewed.

REPORT 3: SCIENTIFIC CLAIMS AND GENERALIZATION

STATUS: PROVISIONAL — VALID ONLY AFTER FLAGS BELOW ARE RESOLVED

Reason: Prompt 1 returned a verdict of Reject / Fundamental Flaw with Gate = MISMATCH, so this is a map of the destination, not a pass. The specific flags that must close before any section of this narrative becomes real: Flag 1 (self-referential ground truth in the RAG evaluation), Flag 2 (statistically impossible seed-42-vs-aggregate F1 mismatch), Flag 3 (Methods-vs-Results test-split count mismatch), Flag 6 (RAGAS Faithfulness explanation inconsistent with stated methodology), and Flag 8 (Gate under-defended — missing perturbation analysis and explicit limitations statement). Every section below is labeled PROVISIONAL for this reason and none of it should be treated as final manuscript language until those five items are resolved and re-reviewed.

CLAIM-TO-EVIDENCE MISMATCH VERDICT

Verdict: Mismatch confirmed, in the same direction Prompt 1 flagged. The title itself is appropriately bounded — "in a Bandwidth-Constrained Ghanaian University Environment" is honest, local-scale language and should stay exactly as it is. The problem sits one level down, in the Objectives, RQs, and Results prose, which repeatedly use unqualified, general-sounding language — "confirming that the weighted cross-entropy loss function... is the sole source of the performance improvement," "the system achieves a perfect BERTScore of 1.000" — as if these were portable findings about WCE and CRAG in general, when the underlying evidence is (a) currently invalid due to the self-referential scoring issue, and (b) even once fixed, will still be a single-institution, 145-training-example result with no perturbation testing and no external data.

Correction: Keep the title as-is. Reframe every results-level claim to be explicitly bounded — "within the KNUST Computer Science query distribution studied here" or equivalent — until the Tier 1 requirements (perturbation analysis, explicit limitations statement) are satisfied, and until the RAG metrics are regenerated against an independent gold-standard reference rather than CRAG's own output. Corrected objective-statement example: "To evaluate whether replacing standard cross-entropy with weighted cross-entropy improves out-of-scope F1 and Macro F1 within this institution's query distribution, and to test the robustness of this improvement under paraphrased and dialect-variant query perturbations." Note: this mismatch check will need to be re-run once Prompt 1/2 flags close, since the actual numbers behind these claims may change materially, not just cosmetically.

TIER CONFIRMATION AND JUSTIFICATION

Data-handling procedure used: Neither Option 1 nor Option 2. There is no public or independent benchmark and no separately-tuned synthetic dataset kept apart from the field data — everything currently in the paper (145-example classifier training set, 10,865-chunk lecture corpus, 100-query RAG evaluation set) is drawn from the same single institutional source, and the RAG metrics are additionally self-referential rather than externally anchored.

Given CLAIM TYPE from Prompt 1 (locally-scoped per the title, mismatched in the prose) and this data-handling reality, the only defensible target right now is TIER 1 — The Rigorous Bounded Claim. Tier 2 and Tier 3 both require a genuine external or synthetic-but-separate dataset that this study does not yet have; reaching for either tier before that data exists would itself create a new mismatch. The claim Tier 1 supports is narrower and more honest than what's currently written: that the WCE engineering operation and the CRAG pipeline were built and validated with a rigorous, reproducible methodology for this specific institutional context — not that they generalize.

Model-family override applied: this is a RAG system, so the evidentiary bar is QA/hallucination-benchmark performance (RAGAS Faithfulness, human-gold-standard QA comparison), not accuracy/F1. The paper already reaches for the right metric family — the problem is that the reference standard behind those metrics is currently invalid (Flag 1) and the Faithfulness result is currently mis-explained (Flag 6), not that the wrong metrics were chosen.

Tier 1 requirement checklist, as it stands today:

- Leave-one-site-out CV: not applicable in the literal sense (single institution, single site) — substitute is a cross-cohort or cross-semester holdout once more than one semester of KNUST data is available; note this as a named future milestone rather than pretending it's satisfied.
- Temporal holdout: not currently present; worth adding once the IDL Whisper-transcribed corpus (already planned per Methods 2.3.3) creates a second, later-collected data source.
- Multi-seed/bootstrap: present and adequate (20 seeds) — this is a real strength, once Flag 2's internal consistency issue is resolved.
- Sensitivity/perturbation analysis: missing (Flag 8) — this is the single most important addition for Tier 1 to be genuinely earned.
- Explicit limitations statement: missing (Flag 8) — required alongside, not instead of, the perturbation analysis.

KILL YOUR OWN DARLINGS PARAGRAPH (drafted, ready to insert into Discussion — provisional until Flags 1, 2, 6, 8 close)

"Two results in this study deserve explicit scrutiny rather than quiet acceptance. First, the Identification intent class was trained on only five examples total, and while the weighted cross-entropy formulation recovers this class from complete collapse ($F1 = 0.00$ under standard cross-entropy) to strong performance under WCE, this recovery rests on a training signal so small that a handful of additional or differently-phrased Identification queries could plausibly shift the result substantially; we treat this as an open robustness question rather than a settled win. Second, Nyansapo CRAG scored lower on RAGAS Faithfulness than the Naive RAG baseline in our evaluation, a finding that runs counter to the paper's central hallucination-reduction thesis. Rather than attribute this to a scoring artifact, we treat it as a genuine signal that the CRAG evaluator's query-rewriting step may, in some fraction of cases, retrieve semantically related but lexically distinct chunks that reduce measured groundedness even when answer quality is preserved; this is examined further below and flagged as a concrete direction for threshold retuning rather than dismissed."

SECTION-BY-SECTION WRITING DIRECTIVES (PROVISIONAL — Tier 1 framing)

Introduction

- State the gap (no prior corrective, offline-capable RAG deployment study in a bandwidth-constrained Ghanaian university) exactly as currently written — this part is sound and doesn't need Tier-language changes.
- Close the introduction's contribution statement with Tier 1 language: "we present a rigorously validated, reproducible methodology for building and evaluating this system within the KNUST context," not "we demonstrate that this architecture improves hallucination reduction" without qualification.
- Do not claim the WCE weighting scheme or the CRAG configuration as generalizable findings in this section — save any generalization language for Limitations/Future Work, where it belongs as a stated direction, not a result.

Methodology

- Add one explicit paragraph describing the data-separation reality plainly: single-institution field data only, no external or synthetic benchmark currently merged or compared, so no leakage-across-source risk exists but also no cross-context validation has been performed.
- Once Flag 1 is resolved, describe the gold-standard reference procedure precisely: which 200 human-annotated pairs were used as ground truth for which metrics, and confirm no system's own output was ever used as a reference for itself.
- Add the perturbation-analysis protocol as a named methodological step (paraphrase and Ghanaian-English/Twi-influenced query variants), described with the same rigor as the seed-based evaluation.

Results

- Local Baseline: report the corrected, internally consistent classifier and RAG numbers (post Flags 1, 2, 3) exactly as measured, including any less dramatic margins than currently reported.

- Generalization Stress Test: report the perturbation-analysis results plainly, including any performance drop — do not soften or bury a drop in a subordinate clause.
- Do not include a Tier 3-style "Adaptation Cycle" subsection — there is no external benchmark yet, so this section does not apply at this stage.

Discussion

- Insert the Kill Your Own Darlings paragraph above, in full, without softening either finding.
- Explain the data mechanics behind the RAGAS Faithfulness result using the manual-inspection diagnostic from Prompt 2's Flag 6 fix, not a scorer-artifact explanation.
- Explicitly name what would be needed to move from Tier 1 to Tier 2 (see Forward-Looking Milestones below) as the paper's own stated next step, not left implicit.

Appendix

- Tier 1 does not mandate the full Tier 2/3 appendix suite (PSI/KL-divergence shift analysis, bidirectional adaptation log) since there is no external dataset to compare against yet.
- Include instead: the perturbation-analysis query set and results in full, the corrected per-seed run log underlying Tables 4 and 5, the gold-standard annotation protocol and Cohen's Kappa value, and a Novelty Defensibility Checklist mapping every claim in the abstract to the exact table or figure that supports it — this last item is worth doing even at Tier 1, since it's exactly what a reviewer will attempt to reconstruct manually if you don't do it first.

FORWARD-LOOKING MILESTONES

Tier 1 to Tier 2 — the specific missing ingredient: a genuinely separate dataset, either a public educational-QA or intent-classification benchmark (kept fully apart from the KNUST data and tuned independently, not merged in), or a rigorously disclosed synthetic-query-generation set built to probe out-of-institution phrasing patterns. Nothing else about the current pipeline needs to change to reach Tier 2 — the missing piece is purely a second, independent dataset run through the same engineered CRAG/WCE pipeline without re-tuning to it.

Tier 2 to Tier 3 — the specific missing step: bidirectional validation, meaning the pipeline is trained and tested in both directions (KNUST-to-external and external-to-KNUST), and when performance drops in either direction, the six-step diagnostic cycle is executed and documented as a formal methodological contribution in its own right, not just reported as a number.

Beyond Tier 3 — one suggestion, still within Q1: a multi-institution consortium study across several sub-Saharan African universities running the same offline-capable CRAG architecture, using the adaptation-cycle methodology from Tier 3 as the shared protocol across sites. This would convert Nyansapo from a single-institution deployment story into the founding methodology for a genuinely regional, SDG-4-scale RAG deployment research program — a considerably stronger and more citable contribution than any single-paper claim could be.

MANUSCRIPT-READINESS VERDICT: Provisional Only — Resolve Prompt 1/2 Flags First.

FINAL EDITOR'S NOTE

The bounded, honest version of this claim is genuinely publishable and arguably more interesting than the overreaching one currently on the page, so there is nothing to lose and everything to gain by fixing the five flags before writing another word of this narrative as final.

ADDENDUM — CONSOLIDATED STUDENT PACKET (Group 1, Nyansapo)

1. WHAT TO FIX FIRST (Fatal/Major flags from Prompt 1, in order)
2. Self-referential ground truth in the RAG evaluation (Fatal) — CRAG's outputs were used as the reference standard for BERTScore, ROUGE-L, the 5-fold CV, and the topic-fairness table, guaranteeing perfect scores by construction.
3. Statistically impossible per-seed vs. aggregate F1 mismatch (Fatal) — Table 4's seed-42 Macro F1 of 0.67 is roughly ten standard deviations from Table 5's claimed 20-seed mean of 0.1802.

4. Methods-vs-Results test-split count mismatch (Major) — Methods states a 22-example test fold with 0 Identification examples; Results reports 29 examples with 1 Identification example.
5. Narrative text contradicts its own results table (Major) — text claims Definition F1 = 1.00, table shows 0.86.
6. Zero-variance "too perfect" CRAG scores across folds (Major) — same root cause as item 1, requires the same fix plus confirmation that real variance reappears.
7. RAGAS Faithfulness result contradicted by an inconsistent post-hoc explanation (Major) — CRAG's lower Faithfulness score versus Naive RAG is explained using a mechanism (keyword-overlap scorer) that contradicts the Methods section's own description of RAGAS as an LLM-judged metric.
8. Gate under-defended: locally-scoped claim missing internal rigor (Major) — no perturbation/sensitivity analysis, no explicit limitations statement scoping the WCE/CRAG claims.
9. Field validation dataset promised in Methods, absent from Results (Minor) — polish item, address after the above.
10. HOW TO FIX EACH ONE

Flag 1 — Rebuild BERTScore, ROUGE-L, the 5-fold CV, and the topic-fairness table using the 200 human-annotated gold-standard QA pairs (Methods 2.3.3) as the reference, never CRAG's own generated answers.

```
your_gold_reference_answers = load_gold_standard_qa_pairs()
for system_name, outputs in your_system_outputs.items():
    bertscore_result = bert_score.score(cands=outputs, refs=your_gold_reference_answers, lang="en")
    rouge_result = rouge_scorer.score_batch(outputs, your_gold_reference_answers)
```

Rewrite 3.2.2 to state explicitly that scoring was against the independent gold set, and carry the corrected (non-1.000) values through Tables 7, 8, 9, 12, and 14.

Flag 2 — Trace Table 4 and Table 5 to one canonical run log and check consistency before writing either table.

```
seed_42_macro_f1 = your_seed_log.loc[your_seed_log["seed"] == 42, "macro_f1"].values[0]
z_score = (seed_42_macro_f1 - your_seed_log["macro_f1"].mean()) / your_seed_log["macro_f1"].std()
assert abs(z_score) < 3, "Seed 42 result inconsistent with aggregate — resolve before writing this table."
```

Rewrite 3.1.1 to describe Table 4 as "a representative seeded run" only once its value is confirmed plausible under the aggregate distribution.

Flag 3 — Regenerate the actual split from the code and confirm Methods Table 2 and Results Table 4 report identical counts.

```
your_train_df, your_temp_df = train_test_split(your_intent_dataset, test_size=0.30,
stratify=your_intent_dataset["intent_label"], random_state=42)
your_val_df, your_test_df = train_test_split(your_temp_df, test_size=0.50, stratify=your_temp_df["intent_label"],
random_state=42)
print(your_test_df["intent_label"].value_counts())
```

If the split logic genuinely changed to guarantee at least one Identification test example, state that explicitly in 2.3.1; otherwise fix the split so both tables match.

Flag 4 — Generate every quoted prose number directly from its source table rather than typing it from memory.

```
definition_f1 = your_classification_report_df.loc["Definition (Class 0)", "F1-Score"]
print(f"The baseline achieves F1 = {definition_f1:.2f} for the dominant Definition class.")
```

Flag 5 — Same code fix as Flag 1, applied to the 5-fold loop; report the resulting variance honestly rather than treating non-zero SD as a problem.

```
for fold_seed in [42, 123, 456, 789, 1024]:
    your_fold_test_set = get_fold(your_gold_reference_answers, seed=fold_seed)
```

```
score = rouge_scorer.score_batch(generate_answers(nyansapo_crag_pipeline, your_fold_test_set["queries"]),
your_fold_test_set["gold_answers"])
```

Flag 6 — Manually inspect low-Faithfulness CRAG cases to find the real cause before writing any explanation.

```
low_faithfulness_cases = your_ragas_results_df[(your_ragas_results_df["system"] == "crag") &
(your_ragas_results_df["faithfulness"] < 0.5)]
for _, row in low_faithfulness_cases.head(10).iterrows():
print(row["original_query"], row["rewritten_query"], row["retrieved_chunks"], row["generated_answer"])
```

Rewrite the Faithfulness paragraph to report the actual diagnosed cause (e.g., query-rewriting shifting retrieved vocabulary) as a genuine, disclosed limitation.

Flag 7 — Run a perturbation analysis (paraphrase and Ghanaian-English/Twi-influenced query variants) and add an explicit limitations sentence scoping the WCE/CRAG claim to this institution's data.

```
your_perturbed_queries = [paraphrase_or_lightly_reword(q) for q in your_test_queries] +
[add_ghanaian_english_phrasing(q) for q in your_test_queries]
perturbed_macro_f1 = evaluate(nyansapo_qic_wce_model, your_perturbed_queries, your_true_labels)
```

Add to Limitations: "Generalizability of the specific class weights and the WCE formulation to other institutions, disciplines, or query distributions has not been established and is a direction for future work."

Flag 8 (Minor) — Either report the field-validation Macro F1 and out-of-scope F1 from the June 11–20 study, or add one sentence in Limitations stating that data collection had not concluded at the time of this draft, with an expected completion date.

3. WHAT THE FINISHED PAPER SHOULD LOOK LIKE (PROVISIONAL)

Tier: Tier 1 — The Rigorous Bounded Claim, provisional until Flags 1–7 above are resolved and re-reviewed. Title stays as-is; Objectives, RQs, and Results prose are reframed to explicitly bound every claim to the KNUST query distribution rather than implying general portability. Methodology adds an explicit data-separation statement and the perturbation-analysis protocol. Results reports the corrected local baseline and the perturbation stress test honestly, including any drop. Discussion includes the Kill Your Own Darlings paragraph on the Identification-class training size and the CRAG Faithfulness finding. Appendix includes the perturbation query set and results, the corrected per-seed run log, the annotation protocol with Cohen's Kappa, and a Novelty Defensibility Checklist mapping every abstract claim to its supporting table. No Tier 2/3 appendix items (external-benchmark shift analysis, bidirectional adaptation cycle) apply yet, since no external dataset currently exists in this study.

4. RESUBMISSION INSTRUCTION

Once you have made these fixes, submit your revised Methodology, Results, and updated code. This will go through a full fresh review — not just a check against this list — since a fix for one issue can sometimes create a new one elsewhere. Do not adjust wording or formatting to work around a flag; the underlying method must actually change.

REPORT 4: RESULTS AND ANALYSIS SECTION ONLY

STEP 0 — MODEL FAMILY IDENTIFICATION

This draft does not map cleanly onto a single Part E model family, and that itself is a critical clarity issue that must be flagged before scoring can proceed. The Results section covers three distinct research objectives built on two structurally different model types:

RQ1 (Section 3.1) evaluates Nyansapo-QIC-WCE, a feedforward neural network — confirmed by Methods Section 2.2.4's stated architecture: "Input(500) → Linear(256) → ReLU → Dropout(0.3) → Linear(128) → ReLU → Dropout(0.3) → Linear(5) → Softmax." This is a Deep Neural Network / MLP under Part E category B.

RQ2 (Section 3.2) evaluates Nyansapo CRAG, a retrieval-augmented generation pipeline (BM25+dense retrieval, CRAG evaluator, Phi-2/Phi-4-mini generation) — this does not correspond to any single category in Part E's taxonomy at all; it is closest to the RAG-specific override noted in the companion Prompt 3 ("RAG needs QA/hallucination benchmarks, not accuracy/F1"), which is outside this marking scheme's Part E list entirely.

RQ3 (Section 3.3) reports deployment latency, which is an engineering/systems benchmark, not a predictive-model result in the Part E sense at all.

Because Part E's visual taxonomy is defined per single model family, and this draft mixes an MLP classifier with a non-taxonomized RAG pipeline and a systems-latency benchmark, the "model-appropriate visual" checks in Part A (A3, A5, A6, A12) cannot be applied uniformly. I am proceeding with RQ1's Nyansapo-QIC-WCE classifier as the primary model family for the mandatory visual checks (Category B — Deep Neural Networks), since it is the only component with a Part E-defined checklist, and I am separately noting where the RAG-pipeline results fall outside this rubric's scope. This should be corrected before resubmission: either split this into two Results sections scored separately (classifier and RAG system), or the marking scheme needs a RAG-specific addendum.

STEP 1 — FAIL CHECK: PASS OR FAIL

A1. DISCUSSION SMUGGLING — FAIL

- "confirming that the weighted cross-entropy loss function — and specifically the computed class weights — is the sole source of the performance improvement, not the architecture change or training procedure" (Section 3.1.3, Ablation Study) — causal attribution language, belongs in Discussion.
- "reflecting the fact that naïve RAG retrieves more chunks per query (with no quality evaluation step) and thus provides more retrieved context for the Faithfulness scorer to match against" (Section 3.2.5) — causal explanation of a result, belongs in Discussion.
- "This unexpected result reflects the shared resource overhead of the Colab T4 environment and the efficiency of Apple Silicon's Neural Processing Unit for transformer inference without GPU" (Section 3.3) — causal speculation, belongs in Discussion.

- "indicating that ungrounded generation is especially unreliable for algorithmic content that requires precise definitional accuracy" (Section 3.2.7) — generalization/implication, belongs in Discussion.
- "confirming that the system does not systematically disadvantage students in any CS subject domain" (Section 3.2.7) — interpretive claim, belongs in Discussion.
- "confirming the viability of offline deployment at KNUST without requiring internet connectivity or server infrastructure" (Section 3.3) — implication/recommendation-adjacent, belongs in Discussion.

A2. MISSING TABLE 3 (main performance comparison, mean $\pm$ SD) — PASS, with a numbering caveat. Table 5 (Section 3.1.2) serves this function for the classifier, and Tables 7–8 serve it for the RAG system, all reported as mean $\pm$ SD. The numbering runs 4–14 rather than starting at 3 because Methods claimed Tables 1–3; the content requirement is met even though the specific number "Table 3" doesn't appear in Results.

A3. MISSING FIGURE 4 (model-appropriate primary trade-off curve) — FAIL. For the identified DNN classifier, Figure 4 should be a PR/ROC curve. No such curve appears anywhere in Section 3.1 or the Figures Reference List (Section 3.5); the closest item, Figure 2 ("qic_engineering_results.png"), is described only as "Macro F1 bar chart, F1 distribution boxplot, per-seed line chart across 20 seeds" — none of which is a PR/ROC curve.

A4. MISSING FIGURE 5 (CV stability box/violin plot) — PASS, content present but mislabeled. Figure 2's "F1 distribution boxplot" across 20 seeds functionally satisfies this, though it is folded into a three-panel composite figure rather than a standalone, correctly numbered Figure 5.

A5. MISSING FIGURE 6 (model-appropriate interpretability visual) — FAIL. Methods Section 2.2.4 states explicitly that "token-level feature weights are directly inspectable without SHAP," and no substitute interpretability visual (coefficient plot, attention map, or otherwise) appears anywhere in Results or the Figures Reference List. For an MLP, some feature/weight-based interpretability visual is still required; none is provided.

A6. MISSING FIGURE 7 (error-analysis visual) — PASS, content present but mislabeled. Figure 1 ("confusion_matrices.png") provides side-by-side baseline vs. engineered confusion matrices per Section 3.1.4, though it is unclear from the text whether these are row-normalized as the rubric requires, and it is numbered Figure 1 rather than Figure 7.

A7. WRONG TENSE REGISTER (future/proposal tense) — PASS at the Part A gate level. No future-tense sentences ("will achieve," "we will evaluate") were found. Note: the draft does use present tense pervasively for observed outcomes ("achieves," "confirms," "reflects"), which is a Section 11 register violation, but this is distinct from the future/proposal-tense trigger that A7 specifically checks.

A8. "ENGINEERED" TERMINOLOGY ABUSE — FAIL. Direct instances of "engineered" in Results: "the engineered Nyansapo-QIC-WCE" (3.1.2 intro), "The engineered model achieves a mean Macro F1 of 0.8249" (3.1.2), "The engineered model achieves perfect classification" (3.1.2), the Figure 1 caption "(engineered)" (3.1.4), and "the engineered model demonstrates more consistent Out-of-scope detection" (3.1.4) — five direct instances, plus root-form variants ("Model Engineering" as the Section 3.1 heading, "the model engineering evaluation (RQ1)" in the section intro, "RQ1 — Model Engineering" in Table 14). This exceeds the six-instance ceiling with no synonym variation — no instance of "proposed variant," "the WCE classifier," "developed model," or similar substitutes appears anywhere for this component.

A9. MISSING STATISTICAL SIGNIFICANCE — PASS. Wilcoxon signed-rank tests, exact p-values, and Cohen's d are reported throughout Tables 5, 6, 7, and 8.

A10. ORPHAN VISUALS — FAIL. Figures 3 ("results_chart.png"), 4 ("final_results_chart.png"), 5 ("latency_chart.png"), and 6 ("complete_evaluation.png") appear only in the Section 3.5 Figures Reference List at the very end of the document and are never cited inline within the analytical text of Sections 3.2 or 3.3 — for example, no sentence in Section 3.2.2 says "as shown in Figure 3" despite Figure 3 being the BERTScore comparison chart per the reference list. A figure mentioned only in an end-of-section list, with no analytical tie-in where the claim is made, functions as an orphan visual.

A11. NUMERICAL IMPOSSIBILITY — FAIL (flagged as VERIFY, not assumed misconduct, per this item's own instruction).

- "Macro F1" of 0.67 computed from Table 4's per-class values (0.86, 0.75, 0.00, 0.75, 1.00 → average 0.672) for the seed-42 run, versus "a mean Macro F1 of 0.1802 (± 0.0459)" across all 20 seeds including seed 42 (Table 5) — VERIFY: possible data-entry error, these cannot both be correct given the stated standard deviation.
- Table 4's stated test-fold size of $n=29$ versus Methods Section 2.3.1 Table 2's stratified test fold of $n=22$ — VERIFY: possible data-entry error.
- The prose claim "it achieves $F1 = 1.00$ for the dominant Definition class" (Section 3.1.1) versus Table 4's own listed Definition (Class 0) F1-Score of 0.86 — VERIFY: possible data-entry error.
- Table 11's "Hybrid RRF (proposed)" ROUGE-L of 0.633 versus Table 8's "Nyansapo CRAG (proposed)" ROUGE-L of 1.0000 — the same "proposed" system reported at two different ROUGE-L values with no explanation of the discrepancy — VERIFY: possible data-entry error.

A12. MODEL-VISUAL MISMATCH — FAIL. For the identified DNN classifier, Category B unavoidable visuals require training/validation convergence curves (loss vs. epoch) as PRIMARY, non-optional evidence per Part E. No convergence curve appears anywhere in Results or the Figures Reference List; only the confusion matrix and F1-distribution boxplot are offered, which is an incomplete evidentiary set for a DNN.

A13. METHODS VISUAL DUPLICATED IN RESULTS — PASS. No Methods-numbered table or figure (Table 1, Table 2, Table 3, Algorithm 1) is reprinted in the Results section; Results tables begin numbering at Table 4 and no overlap was found.

Per this marking scheme's own protocol: "If any item fails, stop here and do not proceed to scoring." Seven Part A items failed (A1, A3, A5, A8, A10, A11, A12), so Steps 2 through 7 (visual audit, section-by-section scoring, tense/discussion correction pass, gate checks, numerical consistency listing) are not performed on this draft. Scoring against the 100-mark rubric would not be meaningful until these structural failures are resolved, since several of them (A1, A11) invalidate the factual basis Sections 2, 5, and 7 would need to score against.

FINAL DECISION

Model Family Identified: Mixed — Nyansapo-QIC-WCE (Deep Neural Network / MLP, Part E Category B) for RQ1; Nyansapo CRAG (RAG pipeline, outside Part E's defined taxonomy) for RQ2; deployment latency benchmark (outside Part E's taxonomy) for RQ3. This mismatch is itself

a critical clarity issue requiring resolution before the draft can be scored as a single Results section.

Part A: FAIL (A1, A3, A5, A8, A10, A11, A12 triggered)

Part B Subtotal: Not scored — scoring is withheld per protocol until Part A passes.

DECISION: REVISE AND RESUBMIT — below the 90% threshold cannot even be assessed until Part A clears.

Three highest-priority fixes before resubmission:

1. Resolve the numerical inconsistencies flagged under A11 first — the seed-42-vs-aggregate F1 mismatch, the $n=22$ -vs- $n=29$ test-fold discrepancy, and the two different ROUGE-L values both attributed to "the proposed system" (Table 8 vs. Table 11). Nothing else can be meaningfully scored while the underlying numbers contradict each other, since Sections 2, 5, and 7 of this rubric all depend on those exact figures being trustworthy.
2. Strip every Discussion-smuggling phrase identified under A1 out of the Results section and move the underlying causal reasoning into the Discussion section where it belongs — replace each with a purely factual restatement of the number (e.g., replace "reflecting the fact that naïve RAG retrieves more chunks per query... thus provides more retrieved context" with "Naive RAG retrieved a mean of [X] chunks per query, compared with [Y] for CRAG").
3. Decide how this multi-objective paper will be structured for this rubric: either produce the missing DNN-mandatory visuals (training/validation convergence curves for the classifier, a PR/ROC curve, and a genuine interpretability visual) and correctly number and inline-reference all figures per Figure/Table order, or split the Results section into two clearly separated sub-reports (classifier results scored against Part E Category B, RAG results scored under a separate rubric addendum) so that "model-appropriate visual" checks can be applied honestly rather than fudged across two incompatible model families.

REPORT 5: GITHUB CODE ANALYSIS

(NEXT TIME INSERT IN THE FILE AND NOT IN THE EMAIL)

I pulled the actual repository (github.com/maagyei04/Nyansapo-RAG-Research) and read through the full experiment notebook, `Nyansapo_RAG_Pipeline_v1.ipynb` (56 cells, both source and saved outputs). This confirms several of the Prompt 1 flags directly in the code, and surfaces four additional issues that weren't visible from the manuscript alone. I'm citing exact cells so these can be checked independently.

CONFIRMED FROM CODE — flags already raised, now with direct evidence

1. The CRAG self-referential scoring is explicit and intentional in the code, not a manuscript-writing slip.

Cell 34 (BERTScore): `references = crag_answers`, then `crag_scores = get_bertscore(crag_answers, references)` — CRAG is literally scored against itself.

Cell computing ROUGE-L: the comment reads `# Use CRAG as reference` immediately above `references = crag_answers`.

The same pattern recurs in the retrieval-ablation cell (`references = [r.get("crag_answer", "") for r in results]`) and the 5-fold CV cell (`references = [r.get("crag_answer", "") for r in fold]`).

This means every RAG-comparison number in the paper — Tables 7, 8, 9, 11, 12, and 14 — measures "how similar is System X's answer to what CRAG said," not correctness against any ground truth. It is not an isolated error in one table; it is the single evaluation function reused everywhere in Section 3.2. This upgrades the earlier Fatal flag from "identified via internal inconsistency" to "confirmed via source code, present by design across the entire RAG evaluation."

2. The Table 4-vs-Table 5 statistical impossibility is explained by two different code paths, not a typo.

Cell 51 (the actual 20-seed run behind Table 5) prints: `Seed 42: 0.1455` for the baseline. Table 5's aggregate (mean 0.1802 ± 0.0459) is computed from this run.

Cell 54 (the cell that actually produces Table 4's confusion matrix) is a separate, later cell that retrains the baseline from scratch at seed 42 with `epochs=200` instead of the `train_model()` function's default `epochs=100`, and reports a macro F1 of 0.67 for that separate run.

So "seed 42" in Table 4 and "seed 42" in Table 5 are two different trained models, produced by two different training loops with different epoch budgets, run at different points in the notebook. They were never going to match, and they don't. The manuscript presents them as the same thing.

3. Methods' 70/15/15 stratified split does not exist anywhere in the code.

Every `train_test_split` call in the notebook (cell 49's `train_model()`, cell 53, cell 54) uses `test_size=0.2`, `stratify=y` — an 80/20 split with no separate validation partition at all. Methods Table 2's 70/15/15 breakdown (test $n=22$) was never produced by any code in this repository; the actual split that produced Table 4's $n=29$ is the 80/20 split, confirmed by $145 \times 0.2 = 29$.

4. RAGAS Faithfulness is not RAGAS.

Searching the entire 17,000-line notebook for `from ragas import` or

`ragas.metrics` returns zero results. The `ragas` package is `pip install`-ed in the setup cell and never imported again. The actual "Faithfulness" scoring cell prints "Computing RAGAS Faithfulness manually..." and implements a stopword-filtered bag-of-words overlap between the answer and retrieved chunks — a simple keyword-overlap heuristic, not the LLM-as-judge metric that Methods Section 2.5.2 cites (from `ragas import evaluate`, Es et al. 2023). This is worse than Prompt 1's Flag 6 assumed: it isn't that the explanation of the Faithfulness result was inconsistent with the method — the method itself was never run. Every "RAGAS Faithfulness" number and citation in the manuscript describes a library that was installed but not used.

NEW — found only by reading the code, not visible from the manuscript

5. Test-set leakage via epoch-level model selection (this is a new Fatal-level finding). In `train_model()` (cell 49), the training loop evaluates the model on `X_test_t/y_test_t` after every single epoch, labels this step with the comment `# Validation`, and keeps `best_f1/best_state` whenever that epoch's test-set F1 improves, using patience-based early stopping on that same test F1. There is no third, held-out validation partition anywhere — the "validation" set used for early stopping is the identical set later reported as the test result. This means every classifier F1 reported in the paper (Tables 4, 5, 6, and the confusion matrices) is the best epoch chosen by looking at test-set performance, not a result measured after training completed. This is a textbook data-leakage pattern, and it plausibly explains a meaningful share of the WCE "improvement," independent of whether the weighting itself has real value. It also explains why particular seeds (123, 61) hit a perfect 1.0 Macro F1 — the loop searched up to 100–200 epochs for the single best-on-test snapshot.
6. Cherry-picked deployment seed presented without disclosure. Cell 53's own comment: `# Train final model on best seed (seed 123 gave F1=1.0)`. Selecting your best-performing seed to actually deploy is defensible engineering practice, but the manuscript's phrase "the engineered model achieves perfect classification... on 2 of 20 seeds" reads as an incidental observation rather than what it is: the team looked at all 20 outcomes and then built the shipped model from the best one. This should be disclosed explicitly rather than left to look like a spontaneous result.
7. Silent generation failures folded into the "hallucination" comparison. Cell 34's own printed output: `Empty Vanilla answers: 7` (out of 100) — 7% of Vanilla LLM generations produced nothing and were replaced with the literal placeholder string "No answer generated." before being scored. That placeholder then receives whatever BERTScore/ROUGE-L/Faithfulness value a near-empty string gets against a real CRAG answer, which is close to the floor of each metric. Some fraction of "Vanilla LLM hallucinates" in the current numbers is actually "Vanilla LLM sometimes failed to generate a response at all," and these are not the same finding. This isn't disclosed anywhere in the Results section.
8. The 200 human-annotated gold-standard QA pairs and the CHRPE-approved field validation study do not appear anywhere in the codebase. Searching the notebook for "gold," "Kappa," "annotator," "human-annotat," and "CHRPE" returns nothing. The entire RAG evaluation runs on a single 100-item `eval_results.json` (rule-labeled, not human-annotated) — not the 200 human-annotated pairs described in Methods 2.3.3, and not the 5,000-query synthetic scale used throughout the Dataset Size Advisory's statistical-power arguments. This changes the earlier remediation advice: Prompt 2's Flag 1 fix assumed the 200-pair gold set already

existed and just needed to be substituted in as the reference; the code shows it needs to be built from scratch before that fix is even possible.

FORWARD-LOOKING SUGGESTIONS, KEYED TO THE ACTUAL CODE

For Flag 1 (self-reference) and Flag 8 above — build the missing gold set before touching the scoring cells: write a small annotation script that pulls `questions = [r["question"] for r in results]` from `eval_results.json`, and have two CS lecturers independently write or verify a reference answer for each, with the answer stored back under a new key (e.g., `r["gold_answer"]`) rather than reusing `crag_answer`. Only once that key exists should cell 34 and the ROUGE-L cell be changed to `references = [r["gold_answer"] for r in results]`.

For Flag 5 (test-set leakage) — the single highest-priority code fix: modify `train_model()` to carve out a third split before training (`X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, stratify=y, random_state=seed)` then split `X_temp/y_temp` again into validation and test), monitor `best_f1` against the validation split only inside the loop, and compute the reported Macro F1 once, after training stops, on the untouched test split. This is a structural rewrite of the training loop, not a one-line change, and every downstream table (4, 5, 6) needs to be regenerated from it.

For Flag 4 (fabricated RAGAS label) — either actually call the `ragas` library (`from ragas import evaluate, from ragas.metrics import faithfulness`) using an LLM-as-judge backend as Methods 2.5.2 describes, accepting the added inference cost, or rename every instance of "RAGAS Faithfulness" in the manuscript, tables, and README to something accurate like "keyword-overlap groundedness score," and remove the Es et al. (2023) citation from anywhere this heuristic is reported, since that citation currently attaches a peer-reviewed method's name to a different, unpublished heuristic.

For Flag 7 (silent generation failures) — add a line to the BERTScore/Faithfulness cells reporting the failure rate per system explicitly (**Vanilla LLM generation failures: 7/100 (7%)**) directly in the Results table, and consider re-running just the failed 7 queries rather than silently substituting a placeholder string that then gets scored as a real answer.

For Flag 2/3 (inconsistent seed-42 runs, missing 70/15/15 split) — delete cell 54's separate retraining entirely; it is redundant with `train_model()` and is the direct source of the Table 4/Table 5 mismatch. Generate the confusion matrix for Table 4 by calling `train_model()` for seed 42 and reusing its returned `best_preds/y_test` rather than retraining from scratch with a different epoch count. And correct Methods Table 2 to describe the actual 80/20 split used everywhere in the code, rather than leaving a 70/15/15 split in the manuscript that no cell in this repository implements.

FIX AND RESUBMIT A REVISED VERSION (METHOD AND RESULTS) IN SEPARATE FILES FOR VERIFICATION

GROUP 1_ VERIFICATION STEP 2

STEP 2 — REGRESSION SWEEP AND PER-ITEM VERIFICATION

Correction to the derivation printed at step 4, before it propagates. I carried 14 items. The correct figure is **13**. Report 1's Flag 8 ("Gate — under-defended local claim") and the GENERALIZATION GATE: MISMATCH line are the same finding stated twice in the same packet, not two items; scoring both would have inflated the denominator and double-charged the student. They are merged below as **F8**, carried at **weight 5** under the Section 5 rule that a failed GATE is fatal-weight, with the re-tier from the packet's stated Major printed here rather than applied silently. Fatal-weight count is unchanged at 5: F1, F2, F8, G-A, C1.

REGRESSION SWEEP

The revision is a complete rewrite — Methods rebuilt as M1–M21, Results as R1–R10, and a new notebook. Almost nothing survives in a form where "previously correct, now wrong" can apply, so the standing triggers below mostly resolve into checks for newly introduced defects. Each check is named with the trigger that sent me to it.

REGRESSION CHECK — F4 — checked because a rewritten section carries new prose

RESULT: new instance of the same defect. R3 states seed 42's test fold "contained only one Out-of-scope example and no Identification examples". R1 and Table 5 both give Identification $n=1$, and cell 10's saved output at 8b217b4 prints Identification support = 1. Scored under F4.

REGRESSION CHECK — A11 (inside G-A) — checked because every table was regenerated

RESULT: two new numerical errors.

- (i) Table 3 ablated row, Delta column reads +0.0056; R2 prose repeats "+0.006". Manuscript: +0.0056 | computed: $0.1831 - 0.1975 = -0.0144$. Sign and magnitude both wrong. R5 separately describes the same condition as "C: 0.183 vs A: 0.198", i.e. below baseline.
- (ii) R1 survey breakdown: 219 CS + 79 non-CS = 298 correct; 152 male + 144 female = 296 against a stated $n=298$. Scored under G-A.

REGRESSION CHECK — A10 (inside G-A) — checked because figures were renumbered

RESULT: broken cross-reference. M17 sends feature importance to "(Figure 4 in Results)". R6 places feature importance at Figure 6; R2 places PR/ROC curves at Figure 4. Scored under G-A.

REGRESSION CHECK — A2 — checked because tables were renumbered

RESULT: intact. Table 3 carries the main comparison as mean $\pm$ SD.

REGRESSION CHECK — A9 — checked because all statistics were recomputed

RESULT: intact. Wilcoxon, exact p, Cohen's d and bootstrap CI all present.

REGRESSION CHECK — A13 — checked because both sections were rewritten

RESULT: intact. No Methods table or figure is reprinted in Results.

REGRESSION CHECK — R4 preprocessing position — checked because a new commit changed the training path

RESULT: the code path is unchanged from v1 — v1 cell 1 and v1 cell 56 both fit TF-IDF globally before splitting, exactly as v2 cell 8 does. The leakage does not date from this revision. What is new is the Method's assertion that it is absent. Scored under C1.

SILENT WITHDRAWAL REGISTER (Section 1)

Five items present in the previous version, or produced by the student and then omitted. None was scored as compliance.

Gone	Relates to	DAT — what the contribution loses
5-fold cross-validation analysis (old Table 9)	F1	The only stability evidence for the RAG arm. The classifier has 20 seeds; the pipeline now has one pass.
Fairness-by-topic analysis (old Table 12)	F1	The only equity evidence in the study. For an education venue this was the finding a second discipline could use.
Retrieval ablation, hybrid RRF vs components (old Table 11)	G-A / A11	M12 still specifies RRF weights 0.4/0.6 as an engineering decision with nothing left in the manuscript supporting them.
User trust and cognitive load results — section R8, skipped in the numbering	F7	Objective 3 is half-reported, and the Cognitive Load Theory framing in the Theoretical Framework now has no outcome attached to it. figures/Figure9_User_Study.png exists at the read commit and the repository README states 4.18/5.0, n=298, p<0.0001.
Perturbation analysis results	F8	The substitute rigour the gate was failed on. It was run — figures/Figure12_Perturbation_Analysis.png exists and cell 13 printed results — and appears nowhere in the manuscript.

PER-ITEM VERIFICATION — FATAL-WEIGHT FIRST

F1 — Fatal → 5 — Self-referential ground truth in RAG evaluation

EXAMINER: VEX

WAS REQUIRED: Re-run BERTScore, ROUGE-L, the fairness-by-topic and 5-fold-CV analyses against the 200 independent human-annotated gold-standard QA pairs; every table built on CRAG-as-reference discarded **and** regenerated.

WHAT CHANGED: "Loaded 200 gold pairs" then "Scoring against gold answers" — v2 cell 19 saved output at 8b217b4.

RECOMPUTED: every RAG figure in the manuscript reproduces the code's saved output exactly. CRAG ROUGE-L 0.228 ± 0.141 | code 0.2283 ± 0.1410 . Naive 0.224 ± 0.148 | 0.2244 ± 0.1478 . Vanilla 0.208 ± 0.066 | 0.2080 ± 0.0663 . BERTScore CRAG 0.790 ± 0.075 | 0.7897 ± 0.0753 . Vanilla 0.796 ± 0.032 | 0.7963 ± 0.0318 . All p-values and effect sizes match. No divergence to print.

VERDICT: PARTIAL

WOULD OVERTURN: evidence that the answers in gold_qa_pairs.csv were themselves produced by any of the four systems. A reviewer cannot currently check this — the file is not in the repository, only its Drive path is. Separately, v2 cell 15 still sits in the repository stating the RAG tables are pending and self-referential; it is stale and superseded by cells 19–20, but a reviewer reading the notebook top to bottom will hit it first.

SCOPE CHANGE: YES — the fairness-by-topic and 5-fold-CV analyses were discarded and not regenerated.

DAT: the paper loses its only equity result and its only stability estimate for the RAG arm.

STILL MISSING: the fairness-by-topic breakdown across the eight topic areas, and the 5-fold CV, both re-run against the gold answers.

F2 — Fatal → 5 — Seed-42 versus 20-seed aggregate impossibility

EXAMINER: VEX

WAS REQUIRED: Recompute both tables from the same run log and confirm mutual consistency before either is written.

WHAT CHANGED: "Sanity check seed 42: F1=0.1486 z=-0.60  Consistent" — v2 cell 9 saved output at 8b217b4.

RECOMPUTED: Table 5's own per-class baseline values (0.74, 0.00, 0.00, 0.00, 0.00) average to 0.148, matching the printed macro row of 0.15 and the code's 0.1486. Against Table 3's mean of 0.1975 ± 0.0820 that is $z = -0.58$. The engineered seed-42 value of 0.309 sits at $z = -2.00$ against 0.6214 ± 0.1559 . Both tables now originate in one function on one split: cell 9 and cell 10 both call train_model_v2.

VERDICT: UPGRADED

WOULD OVERTURN: any showing that Table 5's seed-42 column and Table 3's aggregate come from different cells or different epoch budgets — the precise failure of the previous version.

BEYOND THE ASK: section R10 was not requested. You built a Non-Confirmatory Results section that reports your *worst* seed (0.309, named as the lowest of 20) and both null RAG comparisons with full numbers. The report asked you to make the tables consistent; you also made the unfavourable results visible in the Results section rather than leaving them for Discussion.

F8 — Gate MISMATCH → 5 — Under-defended local claim

EXAMINER: DAT

RE-TIER PRINTED: the packet states Major; Section 5 verifies a failed GATE at weight 5. Applied at 5 for the whole run.

WAS REQUIRED: an explicit limitations statement scoping the WCE and CRAG claims to this institution and dataset, **and** at minimum a perturbation test on paraphrased and Ghanaian-English-influenced query variants.

WHAT CHANGED: "No claims of transferability to other universities, other disciplines" — M18.

VERDICT: PARTIAL

WOULD OVERTURN: a perturbation table in the Results section computed on correctly aligned query–label pairs. The scoping sentence alone does not close a gate that was failed on two conditions.

STILL MISSING: two things. First, no perturbation result appears anywhere in the manuscript. Second, the perturbation code at the read commit cannot produce a valid one: v2 cell 13 sets `test_q` to the first 22 entries of the shuffled questions list while taking `test_lb1` from the actual stratified test split, so queries and labels are not paired. That misalignment is why the printed baseline reads Macro F1 = 0.1278 on *unperturbed* queries against a reported engineered mean of 0.6214 — the harness is measuring predictions against unrelated labels, and none of its four numbers can support or refute a robustness claim.

G-A — Failed GATE → 5 — Report 4 Part A, seven checks failed

EXAMINER: JOINT

WAS REQUIRED: all thirteen Part A checks at PASS before the draft can be scored at all.

STATUS OF THE SEVEN FAILED CHECKS:

- **A1 Discussion smuggling — NOT CLEARED.** The exact phrase the report quoted survives: R5, "*was the sole source of the performance improvement*". Also R2 "*confirming that the computed inverse-frequency weights... were the sole source*"; R6 "*confirming that the classifier learnt genuine intent boundaries rather than spurious correlations*"; R10 "*This reflected the CRAG query-rewriting step... despite improving retrieval quality on some queries*" — the second half of which is an unevidenced causal claim.
- **A3 PR/ROC curve — CLEARED.** "Figure 4 presents the PR and ROC curves" (R2); figures/Figure4_PR_ROC_Curves.png present at 8b217b4.
- **A5 Interpretability visual — CLEARED.** R6 Figure 6; figures/Figure6_Feature_Importance.png present at 8b217b4.
- **A8 "Engineered" terminology — NOT CLEARED.** 25 instances in the Results section against a six-instance ceiling, with no synonym in use anywhere.
- **A10 Orphan visuals — PART CLEARED.** Figures 4–7 are now cited inline where the claim is made. But Figures 8 and 9 exist at the commit and are cited nowhere; there are no Figures 1–3; and M17's pointer to "Figure 4" contradicts R6's Figure 6.
- **A11 Numerical impossibility — PART CLEARED.** Three of the four original items are resolved. The fourth disappeared with the retrieval ablation rather than being explained. Two new errors introduced: Table 3's ablated Delta (+0.0056 stated, −0.0144 computed) and the survey gender split (296 against n=298).
- **A12 Convergence curves — NOT CLEARED in the manuscript.** Cell 10 prints "*Confusion matrices and convergence curves saved*" and figures/Figure8_Convergence_Curves.png exists at the commit. The required artefact was produced and then not carried into the document.

VERDICT: PARTIAL

WOULD OVERTURN: all seven checks reading PASS. The gate cannot close while A1 and A8 show no substantive change, and three further checks are only part-cleared.

C1 — Newly raised Fatal → 5 — Test-set leakage in model selection

EXAMINER: VEX

WAS REQUIRED: carve out a third split before training; monitor early stopping on validation only; compute the reported Macro F1 once, after training stops, **on the untouched test split**; regenerate every downstream table from it.

WHAT CHANGED: "Monitor VAL only – never test" — v2 cell 7 at 8b217b4. The loop now tracks `best_val_f1`, and test is scored once after the loop breaks. The epoch-selection leakage the report identified is genuinely gone, and the structural rewrite was done as specified.

VERDICT: PARTIAL — the test split is not untouched.

CONTRADICTION: revised Method says "fitted on the training partition only (n=102)"; notebooks/Nyansapo_RAG_Pipeline_v2.ipynb cell 8 at 8b217b4 fits the vectoriser on all 145 questions before any split occurs. VERDICT FOLLOWS THE CODE.

CONTRADICTION: revised Method says "Here $N_{\text{total}} = 102$ (training samples)"; cell 8 computes the weights from $N=145$ — which is what M12's own printed weights require: $145/(5 \times 83) = 0.349$, $145/(5 \times 5) = 5.800$. VERDICT FOLLOWS THE CODE.

This is not an accusation of bad faith and the leakage does not date from this revision — v1 fitted globally too. What is new is M6 and M7 asserting the opposite, M6 citing "code LK-01 requirements" for a control the code does not implement. Students routinely fix a script locally and forget to push; if that is what happened, send the commit.

WOULD OVERTURN: a commit in which `TfidfVectorizer` is fitted inside `train_model_v2` on `X_train` only and transformed onto validation and test, with class weights computed from the training partition, and Tables 3, 4 and 5 regenerated from that run.

STILL MISSING: the vectoriser fit moved inside the split; the class-weight denominator changed to the training count; all three classifier tables re-run.

PER-ITEM VERIFICATION — WEIGHT 3 AND BELOW

F3 — Major → 3 — Methods-versus-Results split count mismatch

EXAMINER: VEX

WAS REQUIRED: regenerate the actual split from the code, report the true per-class counts once, and make Methods and Results reference the identical split.

WHAT CHANGED: "training (70%, n=102), validation (15%, n=21), and test (15%, n=22)" — M9.

RECOMPUTED at 8b217b4: the code's two-stage split (`test_size=0.15`, then `test_size=0.176`) produces **train 101, validation 22, test 22**. Manuscript: 102 / 21 / 22. The test fold and its per-class composition match exactly — Definition 13, Explanation 4, Identification 1, Procedural 3, Out-of-scope 1 — confirming R1 and Table 5 against the code.

VERDICT: PARTIAL

STILL MISSING: train and validation are each off by one. Change M6, M7, M8 and M12 to n=101 training and n=22 validation, or change the split parameters to produce 102/21. Four sentences currently repeat n=102.

F4 — Major → 3 — Narrative contradicts its own table

EXAMINER: VEX

WAS REQUIRED: align narrative claims to exact table values, and add a verification pass comparing every quoted prose number to its source table.

WHAT CHANGED: the flagged instance is gone — no prose claims Definition F1 = 1.00, and Table 5 reports 0.74 baseline / 0.78 engineered with matching text.

VERDICT: PARTIAL

STILL MISSING: the verification pass did not run, and one contradiction survived it. R3 states seed 42's test fold contained "*no Identification examples*"; R1, Table 5 and cell 10's output all give Identification n=1. Delete that clause or correct it to one example.

F5 — Major → 3 — Zero-variance "too perfect" CRAG scores

EXAMINER: VEX

WAS REQUIRED: re-run the fold analysis against the external reference and report genuine variance.

WHAT CHANGED: "Nyansapo CRAG : 0.2283 ± 0.1410 " — v2 cell 19 output at 8b217b4, matching R10's 0.228 ± 0.141 .

VERDICT: FIXED. Non-zero variance is now present on every system and both metrics, and the 1.000 / SD 0.000 results are gone from the manuscript entirely.

F6 — Major → 3 — RAGAS Faithfulness mis-explained and mis-named

EXAMINER: VEX / DAT

WAS REQUIRED: either call the ragas library as Methods described, or rename every instance accurately and remove the Es et al. (2023) citation from the heuristic.

WHAT CHANGED: "A keyword-overlap groundedness score was computed as a secondary metric" — M15. The strings "ragas" and "Es et al" return zero matches anywhere in the revision.

VERDICT: FIXED. The rename route was taken exactly as offered, and R10 now reports the unfavourable result (Naive RAG 0.753 against CRAG 0.591) as a finding rather than a scorer artefact. The causal clause attached to it in R10 is a separate matter and scores under A1, not here.

C2 — Weight 3 by consequence — Cherry-picked deployment seed

EXAMINER: VEX

WAS REQUIRED: disclose explicitly that the shipped model was built from the best of twenty seeds after all outcomes were seen.

WHAT CHANGED: "The trained model weights for the best seed (seed=123) were archived" — M14.

VERDICT: PARTIAL

STILL MISSING: both M14 and R2 call seed 123 "best seed" without stating that the selection was made after all twenty runs completed, on the basis of test Macro F1. R2 then presents seed 123's PR/ROC curves as the model's performance evidence with no caveat. One sentence in M14 closes it.

C3 — Weight 3 by consequence — Undisclosed generation failures

EXAMINER: VEX

WAS REQUIRED: report the per-system generation-failure rate explicitly in the Results table, and consider re-running failed queries rather than scoring a placeholder string.

WHAT CHANGED: nothing. "Empty", "failure rate", "failed to generate" and equivalents return zero matches in the revision. At 8b217b4, cell 19 builds `vanilla_ans` with `.strip().split('\n')[0]` and prints no failure counter, so an empty generation is scored silently as an answer.

VERDICT: NOT FIXED

STILL MISSING: a per-system count of empty or failed generations across the 200 gold queries, printed in Results beside the ROUGE-L and BERTScore means. This matters more now than it did in the previous version, not less: Vanilla LLM currently scores *above* CRAG on BERTScore (0.796 against 0.790) and is statistically indistinguishable on ROUGE-L, so any undisclosed empty outputs move that comparison in the opposite direction from before.

C4 — Weight 3 by consequence — Gold set and field study absent from the codebase

EXAMINER: VEX

WAS REQUIRED: build the 200-pair gold set, which did not exist at the time of the report, before any re-scoring could be done.

WHAT CHANGED: "✅ Loaded 200 gold pairs" with five course names printed — v2 cell 19 output at 8b217b4. The set demonstrably exists and was used.

VERDICT: PARTIAL

STILL MISSING: three things. `gold_qa_pairs.csv` is not committed — only a Google Drive path appears, and M21 offers the file on written request, so no reviewer can verify the reference standard the whole RAG arm now rests on. The user-trust study appears in no notebook cell at the read commit despite its result appearing in the

repository README. And the annotation protocol has been lowered without comment: the original Methods promised Cohen's Kappa ≥ 0.80 between two CS lecturers; M4 now reports two team members reviewing 20% of pairs with 93.3% raw agreement, and "Kappa" appears nowhere in the revision.

F7 — Minor → 1 — Field validation promised in Methods, absent from Results

EXAMINER: VEX

WAS REQUIRED: either report the field-validation Macro F1 and out-of-scope F1, or add one sentence stating that collection had not concluded, with an expected date.

WHAT CHANGED: M4 now describes the June 11–20 study in full and R1 states 298 complete responses — but neither route was taken. No results, no limitation sentence.

VERDICT: NOT FIXED

SCOPE CHANGE: YES — section R8 is skipped in the numbering, figures/Figure9_User_Study.png exists at the commit, and the repository README carries the headline figure. No item asked for its removal. *DAT: this is the only evidence in the study that speaks to the education discipline rather than to the pipeline; without it, Objective 3 is half-reported and the Cognitive Load Theory framing has no outcome attached to it.*

STILL MISSING: section R8 reporting the trust and NASA-TLX results with n, means and dispersion — or one sentence stating they are withheld, and why.

STEP 3 — FATAL STATUS, SCORECARD, SCORE AND VERDICT

1. FATAL STATUS — read this block first

Five items carry fatal weight in this run. Four are outstanding.

- F1 Self-referential ground truth in RAG evaluation PARTIAL
- F2 Seed-42 versus 20-seed aggregate impossibility UPGRADED — closed
- F8 Generalization gate MISMATCH, under-defended local claim ... PARTIAL
- G-A Report 4 Part A gate, seven checks failed PARTIAL
- C1 Test-set leakage in model selection PARTIAL

No fatal item returned NOT FIXED. **FATAL ESCALATION does not trigger** at this round: Section 14 fires only where an item issued a forward-fix block in the previous round comes back NOT FIXED, and all four outstanding fatals show substantive, verified movement.

2. SCORECARD

ID	Tier → weight	Examiner	Verdict	Earned
F1	Fatal → 5	VEX	PARTIAL	2.5
F2	Fatal → 5	VEX	UPGRADED	5.0
F8	Gate MISMATCH → 5	DAT	PARTIAL	2.5
G-A	Failed GATE → 5	JOINT	PARTIAL	2.5
C1	New Fatal → 5	VEX	PARTIAL	2.5
F3	Major → 3	VEX	PARTIAL	1.5
F4	Major → 3	VEX	PARTIAL	1.5
F5	Major → 3	VEX	FIXED	3.0
F6	Major → 3	VEX / DAT	FIXED	3.0
C2	Consequence → 3	VEX	PARTIAL	1.5
C3	Consequence → 3	VEX	NOT FIXED	0.0
C4	Consequence → 3	VEX	PARTIAL	1.5
F7	Minor → 1	VEX	NOT FIXED	0.0

Carried forward from a previous verifier round: **none**. This is the first verification of this revision, so no item was excluded from re-examination. Withdrawn from the denominator: **none** — no item was overridden before issue, and the tier gap is MATCHED, so nothing was raised against a standard the evidence cannot carry.

3. SCORE LINE

WEIGHT 5 (5 items) available 25 earned 15.0
 F1 2.5 + F2 5.0 + F8 2.5 + G-A 2.5 + C1 2.5

WEIGHT 3 (7 items) available 21 earned 12.0
 F3 1.5 + F4 1.5 + F5 3.0 + F6 3.0
 + C2 1.5 + C3 0.0 + C4 1.5

WEIGHT 1 (1 item) available 1 earned 0.0
 F7 0.0

TOTAL available 47 earned 27.0

COMPLIANCE SCORE = 27.0 / 47 = 57.4%

CLEARANCE: NOT APPLICABLE. The originating report ran no C1–C6 null clearance certificate, so there is none to re-score. One line for the supervisor, because it changes what the next round must carry: the revision now contains **genuine null results that did not exist when the report was written** — CRAG versus Naive RAG at $p=0.554$, $d=0.026$, and CRAG versus Vanilla LLM at $p=0.056$ and $p=0.325$. The previous version's RAG comparisons were self-referential and returned 1.000, so no null was available to certify. These nulls have never been through a clearance certificate at any point in this work's history.

4. VERDICT LINE

NOT CLEARED — 57.4% weighted. Outstanding: F1, F8, G-A, C1, F3, F4, C2, C3, C4, F7. Revise and resubmit.

The three clearance conditions, stated with the actual figures:

- **CONDITION 1 — FAILS.** 57.4% against a threshold of 90%.
- **CONDITION 2 — FAILS.** Four fatal-weight items outstanding: F1, F8, G-A, C1. No regression. No newly introduced fatal — the TF-IDF fit position under C1 pre-dates this revision in the code; what is new is the Method's claim that it is absent.
- **CONDITION 3 — WITHHELD.** DAT does not sign. The tier gap is MATCHED and does not withhold the licence on its own, but two headline claims are currently stated above the strength their evidence carries. The RAG comparison claims rest on a reference file no reviewer can reach, and the "sole source of the performance improvement" claim in R2 and R5 rests on an ablation run inside a feature space fitted on all 145 examples including the test fold.

The release decision is not mine. This line is the instrument's arithmetic; whether the work goes back to the student, and on what timetable, belongs to the supervisor at step 18.

5. OUTSTANDING ITEMS — the student's entire next task

Ordered by weight. Each line names the artefact that closes the item, not the argument for it.

Weight 5

1. **F1** — the fairness-by-topic breakdown across the eight topic areas, and the 5-fold CV, both re-run against the 200 gold answers and reported as tables.
2. **F8** — a perturbation table in Results, computed on correctly paired query-label data. Cell 13 currently takes queries from the first 22 entries of the shuffled list and labels from the stratified test split; those are not the same 22 items.
3. **G-A** — five of seven Part A checks still open: strip the causal clauses from R2, R5, R6 and R10 (A1); reduce "engineered" from 25 instances to six with synonyms (A8); cite or remove Figures 8 and 9 and fix M17's pointer from Figure 4 to Figure 6 (A10); correct Table 3's ablated Delta from +0.0056 to -0.0144 and reconcile the 296/298 gender split (A11); place the convergence curves in the manuscript (A12).
4. **C1** — move the `TfidfVectorizer` fit inside `train_model_v2` so it fits on `X_train` and transforms validation and test; compute class weights from the training count; regenerate Tables 3, 4 and 5 from that run and push the commit.

Weight 3

5. **F3** — correct M6, M7, M8 and M12 to training $n=101$ and validation $n=22$, or change the split parameters to produce 102 and 21. The test fold of 22 is already correct and matches the code exactly.
6. **F4** — delete or correct the clause in R3 reading "no Identification examples"; R1, Table 5 and the code all give Identification $n=1$ for that fold.
7. **C2** — one sentence in M14 stating the deployment model was selected after all twenty runs completed, on the basis of test Macro F1.
8. **C3** — a per-system count of empty or failed generations across the 200 gold queries, printed in Results beside the ROUGE-L and BERTScore means.
9. **C4** — commit `gold_qa_pairs.csv` to the repository or resolve the Zenodo DOI; add the user-study analysis cell; and state plainly in M4 that agreement was measured as 93.3% raw agreement on 20% of pairs by two team members, rather than the Cohen's Kappa ≥ 0.80 between two CS lecturers the original Methods promised.

Weight 1

10. **F7** — section R8 reporting the trust and NASA-TLX results with n , means and dispersion, or one sentence stating they are withheld and why. The section number is currently skipped, and the figure and headline result both exist in the repository.

H STEP 4 — FORWARD-FIX BLOCKS

Four fatal-weight items are outstanding and each gets a block below. F2 (the seed-42 versus aggregate impossibility) closed at UPGRADED and needs nothing further.

Do them in the order printed. C1 comes first because re-running it changes the classifier numbers that Tables 3, 4 and 5 report, and three of the Part A checks under G-A are corrections *to those tables*. If you fix the tables first and the split second, you will do the table work twice.

FORWARD-FIX — C1 • Test-set leakage in model selection • weight 5 • PARTIAL

FF1 — THE ONE ACTION. Move the `TfidfVectorizer` fit inside `train_model_v2` in `Nyansapo_RAG_Pipeline_v2.ipynb` so it fits on `X_train` only and transforms validation and test, then re-run cell 9.

FF2 — THE ARTEFACT THAT PROVES IT. A pushed commit hash in which cell 8 passes raw questions and labels into `train_model_v2` rather than a pre-vectorised `X`; the vectoriser is fitted after the two splits; `class_weights_tensor` is computed from the training partition count rather than from 145; and Tables 3, 4 and 5 in the manuscript carry the numbers that run produces. Put the hash in M21 so it can be checked.

FF3 — THE SELF-TEST. Print `len(vectorizer.vocabulary_)` inside the function, then print it again after fitting on all 145. If the two numbers are identical, the fit has not actually moved. Then take one test-fold query, check whether any of its distinctive bigrams appear in the vocabulary, and ask whether that bigram could have entered the vocabulary from the training set alone.

FF4 — TIME COST AND WINDOW. 2–4 hours for the code change, plus roughly 10 minutes of compute for the 20-seed re-run. **FIXABLE AFTER SUBMISSION: NO.**

"This is fixable now and not after submission. Additional collection or finer-resolution collection closes it while your ethics approval and site access are live. Once the manuscript is submitted this becomes a permanent limitation."

That paragraph applies here because the repair cuts your effective n further. Honest splitting already leaves you 101 training and 22 test examples, with a single Identification case in the test fold. Refitting the vectoriser on 101 examples instead of 145

will shrink the feature space again. More labelled queries collected now is the only thing that offsets it, and your CHRPE approval and student access are live today.

FF5 — IF IT CANNOT BE REPAIRED. It can be repaired, so there is no DECLARE exit here, and repair is the only route. One thing to prepare for: the reported 0.6214 mean Macro F1 may fall when the vectoriser stops seeing test queries. If it does, that lower number is your result. A drop is not a reason to reach for DECLARE — that exit exists for defects that cannot be fixed, never for a fix whose outcome you would rather not report.

FF6 — MANUSCRIPT CONSEQUENCE. While this stands, no classifier performance number may be drafted into an abstract, and the ablation conclusion in R2 and R5 — that the computed weights are the sole source of the improvement — may not be written, because the ablation was run inside a feature space built from the full dataset. M6's sentence citing "code LK-01 requirements" must come out or become true; as written it asserts a control the code does not implement.

FORWARD-FIX — F1 • Self-referential ground truth in RAG evaluation • weight 5 • PARTIAL

FF1 — THE ONE ACTION. Add a cell that scores ROUGE-L and BERTScore per topic area and per CV fold against gold_answers, reusing the four answer lists cell 19 already produced, and put both results in the manuscript as tables.

FF2 — THE ARTEFACT THAT PROVES IT. Two tables in Results: a fairness-by-topic table with mean and SD for all four systems across the eight topic areas, showing n per topic; and a 5-fold cross-validation table with mean and SD per fold. Both against the gold answers, neither against any system's own output.

FF3 — THE SELF-TEST. Before you write either table, print the SD of every cell. If any SD is 0.000, the reference has slipped back to a system's own generations — check what variable you passed as refs. Then print the per-topic counts and confirm they sum to 200.

FF4 — TIME COST AND WINDOW. 2–3 days. **FIXABLE AFTER SUBMISSION: YES in principle** — the answers are already generated and the analysis runs on existing data. But the claims these tables support cannot be written until they exist, so deferring them means drafting your Results and Discussion around numbers you have not seen, then rewriting both when they arrive.

One caution on power rather than a window: 200 pairs across eight topics is roughly 25 per topic, which is thin for a per-topic comparison. Report the n beside every topic row so a reviewer can see it, and do not read a topic-level difference as a finding at that size.

FF5 — IF IT CANNOT BE REPAIRED. It can be repaired and there is no DECLARE exit. The gold answers exist, the system outputs exist, and the scoring functions are already written in your notebook.

FF6 — MANUSCRIPT CONSEQUENCE. No fairness or equity claim may be drafted anywhere in the manuscript while the topic breakdown is absent — that includes any sentence saying the system does not disadvantage students in any subject area. No stability claim may be made for the RAG arm: the classifier has 20 seeds, the pipeline currently has one pass, and those cannot be described in the same language.

FORWARD-FIX — F8 • Generalization gate MISMATCH • weight 5 • PARTIAL

FF1 — THE ONE ACTION. Fix the pairing bug in cell 13 — take the perturbation queries from the same indices as the test labels rather than from the first 22 entries of the shuffled list — then re-run it and put the results in Results.

FF2 — THE ARTEFACT THAT PROVES IT. A perturbation table in the Results section with four rows (original, paraphrased, Ghanaian-English, typographic) giving Macro F1 and the drop from original, plus the query count each was computed on, and Figure12_Perturbation_Analysis.png cited inline beside it.

FF3 — THE SELF-TEST. Run the harness on the *unperturbed* test queries first. That number must land close to the engineered model's result for that seed. It currently prints 0.1278 against a reported mean of 0.6214 — that gap is the bug announcing itself, not the model collapsing. Only once the unperturbed row matches do the other three rows mean anything.

FF4 — TIME COST AND WINDOW. 1 day for the fix and re-run. **FIXABLE AFTER SUBMISSION: NO.**

"This is fixable now and not after submission. Additional collection or finer-resolution collection closes it while your ethics approval and site access are live. Once the manuscript is submitted this becomes a permanent limitation."

This applies twice over. Your Ghanaian-English condition is currently generated by prepending stock phrases from a four-item list, which is a reasonable stand-in but not the real thing; real code-switched queries from KNUST students would make the condition genuinely evidential, and you can collect those now. And the Identification class has five examples in total — a granularity fixed at collection time. Collect more Identification queries while you can.

FF5 — IF IT CANNOT BE REPAIRED. The perturbation test is repairable, so no DECLARE exit applies to it. If the repaired test shows the improvement drops sharply under paraphrase, that is a finding to report, not a defect to declare — and it is exactly the question the gate was asking.

FF6 — MANUSCRIPT CONSEQUENCE. M18 currently lists perturbation analysis among the substitute rigour procedures "applied". That sentence cannot stand while no perturbation result appears in Results — either the table goes in or the claim comes out. Until then, no robustness claim may be drafted for the classifier, and the Discussion may not argue that the weighting scheme captures intent rather than surface phrasing.

FORWARD-FIX — G-A • Report 4 Part A gate, five checks open • weight 5 • PARTIAL

FF1 — THE ONE ACTION. Correct the two wrong numbers first: change Table 3's ablated Delta from +0.0056 to -0.0144, and

reconcile the survey gender split, which currently reads 152 male and 144 female against a stated $n=298$.

FF2 — THE ARTEFACT THAT PROVES IT. A revised Results section in which all five open checks read PASS: every causal clause removed from R2, R5, R6 and R10 (A1); "engineered" reduced from 25 instances to six with synonyms such as "the WCE classifier" or "the proposed variant" carrying the rest (A8); Figures 8 and 9 either cited inline or removed, and M17's pointer corrected from "Figure 4" to Figure 6 (A10); both arithmetic errors corrected and R2's "+0.006" prose corrected with them (A11); and the convergence curves placed in the manuscript as a numbered, inline-cited figure (A12).

FF3 — THE SELF-TEST. Two passes you can run alone. First, search the Results section for "confirming", "reflecting", "demonstrating" and "indicating" — every hit is a candidate causal clause that belongs in Discussion. Second, take every number quoted in prose and read it against the table it describes, one at a time, out loud. That second pass is what would have caught R3's claim that seed 42 had no Identification examples when Table 5 beside it shows one.

FF4 — TIME COST AND WINDOW. 1–2 days, all of it writing and figure placement. **FIXABLE AFTER SUBMISSION: NO for A11**, yes for the rest. A wrong number inside a submitted table is read as an integrity problem rather than a revision item, and it is the cheapest thing on this list to fix today.

FF5 — IF IT CANNOT BE REPAIRED. Every one of these is a writing or figure task with no data dependency. There is no DECLARE exit and none is needed; repair is the only route and it is entirely within your control this week.

FF6 — MANUSCRIPT CONSEQUENCE. Part A is a gate, so nothing downstream of it can be scored while it is open — the Results section cannot be marked against the rubric at all, and no section of the manuscript may be drafted from it. Concretely: the abstract may not state the ablation conclusion while A1 stands, and no figure may be referenced in the Discussion that is not first cited in Results.

STEP 5 — COVERAGE, MANUSCRIPT HANDOFF, ENHANCEMENT BLOCK

COVERAGE

ITEMS : 13 of 13 verified, 0 UNVERIFIABLE

CODE RE-READS : 6 performed at 8b217b4eff61cf6b911312661357559f46995c1d,
0 UNVERIFIABLE
R1 seed regime · R2 trial counts · R3 baseline identity
R4 preprocessing position · R5 test-set scoring count
R6 capacity versus n

NOT SEEN : gold_qa_pairs.csv — the reference file the entire RAG arm now rests on. Referenced at M4 and loaded in cell 19 from a Google Drive path; not in the repository, and the Zenodo DOI at M21 reads "pending upload". I verified that 200 pairs loaded and that five course names printed; I could not read a single question or gold answer.

: the user trust and cognitive load analysis. No notebook cell at the read commit produces it, though figures/
Figure9_User_Study.png exists and the repository README states 4.18/5.0, $n=298$, $p<0.0001$.

: CHRPE approval reference number. M5 states it "will be inserted upon receipt of the formal approval document."

: any commit hash in the revised Method. Every code verdict in this run is anchored to a hash you did not supply; if you push before resubmitting, none of them transfers.

One note the supervisor should have in writing. The covering message that accompanied these uploads described a power diagnostic in R10 — achieved power 0.446, $n\geq 411$ for 80% power, $d=0.160$, the ROUGE-L null labelled inconclusive. **No such content exists in the uploaded document.** The strings "power", "inconclusive", "411" and "0.446" return zero matches, and R10's own stated effect size is $d=0.184$. I scored the document that arrived. If that diagnostic exists in a newer draft, it was not the draft submitted, and the difference matters because it is the analysis that would decide whether the $p=0.056$ result is a boundary condition or an underpowered test.

MANUSCRIPT READINESS HANDOFF

CONDITION 2 does not hold — four fatal-weight items are outstanding. Under Section 10's own rule, **only 10E is printed**. The claim licence table, the tier confirmation, the limitations schedule and DAT's line are withheld until F1, F8, G-A and C1 clear. That is not a formality: the licence exists to stop a claim reaching the manuscript unsupported, and three of the four open items bear directly on what the headline claims are allowed to say.

10E — BLOCKED SECTIONS

Blocked	Blocked by	What unblocks it
Abstract — any classifier performance figure	C1	The re-run with the vectoriser fitted inside the split, and Tables 3–5 regenerated from it.
Abstract — the ablation conclusion ("sole source of the improvement")	C1, G-A/A1	The same re-run, plus removal of the causal clauses from R2 and R5.
Discussion — any robustness or transfer argument	F8	A perturbation table in Results computed on correctly paired query–label data.
Discussion — any fairness or equity claim	F1	The fairness-by-topic breakdown across the eight topic areas, re-run against gold answers.
Discussion — any stability claim for the RAG arm	F1	The 5-fold CV against gold answers. The classifier's 20 seeds do not cover the pipeline.
Results — scoring against the Part 4 rubric at all	G-A	All seven Part A checks reading PASS. It is a gate; nothing downstream scores while it is open.
Limitations — the section cannot be finalised	F7, F8, C4	R8 written or its absence stated; the perturbation result; the annotation protocol described as it was actually run.

Everything not listed above is open, and you should start on it today. That includes the entire Introduction, the Problem Statement, the Theoretical Framework, the Related Work survey, and M1 through M5, M10, M11, M19, M20 and M21 — none of which depends on a single outstanding item. The Methods section is in substantially better shape than the Results section, and drafting the framing while the code re-runs is the efficient order.

ENHANCEMENT BLOCK

11A — WHAT YOU DID BEYOND THE ASK

One item was upgraded, and it is worth naming precisely.

The R10 Non-Confirmatory Results section. You were asked to make Table 4 and Table 5 mutually consistent. You did that, and you also built a Results section that reports your *worst* seed — 0.309 at seed 42, named as the lowest of twenty — alongside two null RAG comparisons with full numbers, including the one where the Vanilla LLM scores above your proposed system on BERTScore. No item in the report asked for that section. Reporting a result that runs against your own thesis, in Results rather than buried in Discussion, is the reporting standard the target venue expects and the thing reviewers check first for in this cohort.

Two related moves that sit just under the upgrade line but should not go unremarked: you embedded the z-score consistency assertion into the experiment cell itself, so it runs on every execution rather than being a check someone remembers to perform; and on the RAGAS finding you took the harder of the two routes offered — renaming the metric accurately throughout and removing the Es et al. citation, rather than keeping the borrowed label.

11B — THREE MOVES THAT WOULD LIFT THIS FURTHER

These are optional. They do not affect the compliance score, they are not added to the outstanding list, and none of them is required for clearance.

1. **Report the achieved power on the two nulls.** You have $n=200$, $d=0.184$ and $d=0.026$. Two lines of `statsmodels` turn "not significant" into a statement about what your design could and could not have detected — and it separates the CRAG-versus-Naive result ($d=0.026$, a genuine boundary condition at any

plausible n) from the CRAG-versus-Vanilla result ($d=0.184$ at $p=0.056$, which is a different animal). Reviewers in this domain read an undiagnosed $p=0.056$ as a hedge.

2. **Put the four-way answer set in the repository.** Cell 19 already generates `crag_ans`, `naive_ans`, `bm25_ans` and `vanilla_ans` for all 200 gold questions. Committing those alongside the gold pairs lets a reviewer recompute every RAG number in your paper without a GPU. Few submissions at this level offer that, and it costs you one `json.dump`.
3. **Give the Identification class one honest sentence in Results.** Five examples total, one in the test fold, $F1 = 0.00$ for both models at seed 42. You already report the number; what is missing is the line saying that a single test instance cannot support per-class inference. Naming that yourself is worth more than having a reviewer name it.

11C — ONE LINE OF DIRECTION

The strongest thing in this submission is that your negative results survived the revision — the gold-standard re-run destroyed the perfect scores your previous draft reported, and you published what replaced them rather than reaching for a reframing; protect that while you draft.

CODE LAB ENGINEER SUPPORT

SECTION A3 — NOT MINE

NOT MINE — G-A, F3, F4, C2, F7. Writing and judgement items. They are in your verification report. Nothing below touches them. The withheld claim licence is also theirs, not mine.

SECTION B — OPEN-ISSUE REGISTER

ITEM	CAUSE	CLOSURE	VERDICT	BLOCK	EVIDENCE REQUIRED
C1	11	OPEN IN CODE		V1e -> K1 *	fit located after the split line; weights from the training fold. v2 cell 8 L72 "X = vectorizer. fit_transform(questions)" runs before cell 7 L11 train_test_split. Weights: cell 8 L78 "n_total / (num_classes * class_counts[i])".
C1	4	CLOSES ON RUN		V2	len(X_train), len(X_val), len(X_test) printed from the run. NOT IN SOURCE: any split-size print — v2 cell 7 train_model_v2.
F1	-	CLOSES ON RUN		V4	8 topic rows x 4 systems, mean, SD, n per topic, vs gold_answers. NOT IN SOURCE: fairness / topic analysis — both notebooks, full text search. Depends on C4.
F1	-	CLOSES ON RUN		V1b	5 fold rows x 4 systems, mean and SD per fold, vs gold_answers. NOT IN SOURCE: KFold / n_splits / cross_val — both notebooks.
F8	12	OPEN IN CODE		V3 -> K2 *	pairing overlap count, then four conditions with n per row. v2 cell 13 L86 "test_q = [questions[i] ...][:len(y_test)]" against L87 "test_lbl = list(y_test_t. numpy())". Two different 22 rows.

C3	-	CLOSES ON RUN	V5	empty / failed count per system over 200, as one table row. v2 cell 19 L41 ".strip().split ('\n')[0]" can yield "". No count anywhere in the notebook.
C4	-	OPEN IN CODE	V6	gold_qa_pairs.csv and the four answer lists in the tree at a stated hash. NOT IN SOURCE: any CSV — git ls-files at 8b217b4. BLOCKER: .gitignore L4 "*.json" silently drops the answer lists.
C4	-	CLAIMED, NOT PUSHED	V6	the analysis cell, plus the responses it reads, committed. README publishes "User trust 4.18/5.0 (n=298, p<0.0001)". NOT IN SOURCE: any cell computing it — both notebooks, "nasa tlx trust". Strip identifiers from the n=298 responses before committing.

* K blocks sit behind RUN_CLOSE_BLOCKS = False. Half 1 returns first. (F11)

SECTION C — NULL STATE

NULL STATE: 1 — NO NULL YET | EVIDENCE: no console output returned, and the only p-value in the null-diagnosis cell is a source literal, not a computed value: v2 cell 22 L25 "print(f"Observed p-value : 0.0556")".

You are not in state 5 and the diagnosis cell does not put you there. Cell 22 computes power and required n from gold_rouge_v2.json, but the p-value it prints beside them is typed in. Nothing has returned. One deterministic scoring run over one gold set is one draw.

EXIT ROUTE: V1a-V1f, first in the script, before anything else.

One structural note on V1a, because it changes what the block does. Generation is deterministic at v2 cell 19 L37 do_sample=False, so seeds do not move the RAG arm. Its stability dimension is the evaluation set, and V1a resamples over the 200 gold pairs instead. The classifier arm does vary by seed and cell 9 already runs 20, but those numbers come out of the cause-11 pipeline, so they enter the ledger as a BEFORE row and nothing else.

NO-COMPUTE CONFIRM — answer this in writing, now, before the script is built. Cause 12 is routed on F8 and costs you nothing to settle:

Underline the unit in your hypothesis sentence. Now name the unit your analysis uses. Coarser confirms it. THIS IS FIXABLE NOW AND NOT AFTER SUBMISSION. FURTHER OR FINER COLLECTION CLOSES IT WHILE YOUR ETHICS APPROVAL AND SITE ACCESS ARE LIVE. ONCE THE MANUSCRIPT IS SUBMITTED THIS BECOMES A PERMANENT LIMITATION.

That warning covers cause 4 on C1 and cause 12 on F8. Both are on your handover's not-fixable-after-submission line.

THE SCRIPT BELOW IS AN EXEMPLAR. IT IS NOT A FORCED SOLUTION. RUN IT IN YOUR OWN ENVIRONMENT. CHECK EVERY COLUMN NAME, FILE PATH, OBJECT NAME, SEED, LIBRARY VERSION AND DATA CONDITION. MODIFY IT WHERE YOUR STUDY CONTEXT REQUIRES MODIFICATION. YOU OWN THE FINAL VERSION. DO NOT RUN IT BLINDLY.

IT PRODUCES EVIDENCE, NOT OUTCOMES. IT CAN COME BACK WITH A RESULT YOU DID NOT WANT. THAT IS WHAT MAKES IT WORTH RUNNING. REPORT WHAT RETURNS, INCLUDING WHEN IT CONTRADICTS YOUR DRAFT.

SOME OF THESE CLOSURES WILL MAKE YOUR NUMBER WORSE. THAT IS THE CORRECT OUTCOME FOR SIX OF THE SEVENTEEN CAUSES, AND EVERY BLOCK TELLS YOU IN ADVANCE

WHICH KIND IT IS. A SMALLER GAP AFTER FIXING A LEAK IS THE FIRST HONEST NUMBER YOU HAVE HAD.

THIS IS THE LAST ENGINEERING PASS. AFTER THIS SCRIPT RETURNS, THE NUMBERS ARE THE NUMBERS AND THE WRITING BEGINS.

K1 is CORRECTING. Expect 0.6214 against 0.1975 to shrink, and Wilcoxon p and Cohen's d to move with it. Figures 4 through 8 and Tables 3, 4, 5 are all downstream of cell 8 L72 and all regenerate.

LEDGER — RUN-GROUP_1-20260913
OPEN AT START : 8
CLOSED IN CODE : 0
STILL OPEN : C1 (x2), F1 (x2), F8, C3, C4 (x2)
AWAITING PASTE-BACK: NONE
NULL STATE : 1 — NO NULL YET
COMPUTE ESTIMATED : NOT SET — ceiling not supplied; estimate printed at STEP 2

OPEN IN CODE: 8 of 8

COMPUTE ESTIMATE — against COMPUTE CEILING: NOT SET
Expensive refits : 1 of 5 permitted (V1d only)
LLM generations : 0 — Half 1 re-scores nothing; it reads the arrays
cell 19 and cell 20 already wrote
Bootstrap resamples : 2000 on a 200-length vector — seconds
Wall clock estimate : under 3 minutes on the runtime that produced the
original results
WARNING: no ceiling was supplied, so this estimate is unchecked against one.
If your runtime is tighter than the above, say so before running.

No cause has returned a confirm from your own run. Half 2 is therefore not in this file (F11). Close blocks are emitted at the next turn, one cause at a time, against the table you paste back.

SECTION D — RUN ORDER

RUN THIS FIRST. DO NOT SET RUN_CLOSE_BLOCKS UNTIL THE HALF 1 TABLE IS RETURNED.

1. Open Nyansapo_RAG_Pipeline_v2.ipynb. Run cells 1 to 9 as they stand.
Half 1 reads your live objects. It changes none of them.
2. Fill the four <<< FILL IN >>> constants in CONFIG. They are the only fields this file cannot derive:
DATA_DIR — only if ENV prints "local"
NOTEBOOK_PATH — where v2 .ipynb sits in this session
TOPIC_COL — the column in gold_qa_pairs.csv holding the 8 topics
TLX_RESPONSES — the n=298 response file, or leave as-is if you do not have it
3. Paste the whole file in as the next cell and run it. Do not run it as a separate process — it binds to objects that only exist in your session.
4. Paste back the VERDICT TABLE. The whole table, not the rows you like.

SECTION E — THE CLOSE-OUT SCRIPT

One file, 789 lines, Half 1 only. Syntax checked. Naming scrub returned zero hits.

BLOCK MANIFEST — every block's own header carries its item, cause, expected direction, runtime and paste-output target in the code.

BLOCK	CLOSES	DIRECTION	FITS	WHAT IT DECIDES
V1e	C1 / cause 11	NEUTRAL	0	the cause-11 confirm. Gates K1.
V1f	certificate C4	NEUTRAL	0	tuning parity between arms
V1a	C1 (cert) / null	NEUTRAL	0	sign stability under resampling
V1b	F1 (5-fold)	NEUTRAL	0	does the ranking hold fold to fold
V1c	certificate C3	NEUTRAL	0	MDE, paired solver

V1d	certificate C5	NEUTRAL	1	base rate and per-class recall
V2	C1 / cause 4	NEUTRAL	0	the three fold sizes that ran
V3	F8 / cause 12	NEUTRAL	0	the pairing count. Gates K2.
V4	F1 (fairness)	NEUTRAL	0	per-topic table, 4 systems
V5	C3	NEUTRAL	0	empty generations per system
V6	C4 (both rows)	NEUTRAL	0	what must be committed

Every block is NEUTRAL because Half 1 measures and repairs nothing. The CORRECTING blocks are K1 and K2 and they are not in this file.

SECTION F — CORRECTIONS LEDGER TEMPLATE

Fill a row every time a number moves. Paste the finished table into your appendix.

CAUSE | CONFIRMED BY | FIX APPLIED | BEFORE | AFTER

11			0.6214	
11			0.1975	
11			p=0.000002	
11			d=2.1750	
12				

BEFORE values are QUOTED from your README and Results. They are the starting figures, not calculations.

A reviewer who sees a number move with no explanation suspects the search. A reviewer who sees this table reads a lab with a defect-tracking process. Same movements, opposite conclusion. A number that moved with no row here is reverted. (F12)

SECTION G — CLOSE-OUT CHECKLIST

- [] every block carries its item and cause number
- [] version guard and preflight ran clean, or the assert that fired is reported as a finding
- [] the verdict table is pasted back in full
- [] no column name was invented
- [] every numeric literal carries a source
- [] no data is silently dropped
- [] no result was targeted
- [] every number that moved has a cause row in the ledger
- [] full grids reported, never the best point
- [] unfavourable results disclosed
- [] the certificate is 6 of 6, or the null is not claimed as a finding
- [] every number in the manuscript traces to ONE locked run
- [] the commit hash is stated and postdates every fix
- [] no lab identifier appears anywhere in the code

The last box is not rhetorical for this repository. Two comment lines fail it: v2 cell 0 L3 and cell 13 L4.

LEDGER — RUN-GROUP_1-20260913
OPEN AT START : 8
CLOSED IN CODE : 0
STILL OPEN : C1 (x2), F1 (x2), F8, C3, C4 (x2)
AWAITING PASTE-BACK: V1a V1b V1c V1d V1e V1f V2 V3 V4 V5 V6
NULL STATE : 1 — computed by the script from V1a-V1f
COMPUTE ESTIMATED : 1 refit of 5 permitted, against COMPUTE CEILING NOT SET
NOT CLOSED. STILL OPEN: C1, F1, F8, C3, C4. RUN V1a-V6 AND PASTE THE VERDICT TABLE.
[PAUSE] RUN IT AND PASTE BACK THE VERDICT TABLE.

SEE ATTACHED SCRIPT AND RUN.

GROUP 1_VERIFICATION CODE CHECKS

WHAT GENUINELY CLOSED

Six of eight. These are real fixes and they are correctly done.

ITEM	VERDICT	ANCHOR
C1 (11)	CLOSED IN CODE	cell 7 L11 "X_train = vec.fit_transform(q_train)" with L12-13 transforming val and test. Weights at L23 "n_train / (num_classes * train_counts.get(i,1))". Both halves of the fix landed. This is correct.
C1 (4)	CLOSED	Method corrected to 101/22/22. Cell 10 V2 prints them and does not touch train_model_v2.
F8 (12)	CLOSED IN CODE	cell 17 L41-42 takes model, vectoriser and test fold from ONE run; L53-54 "test_q_fixed = q_test_123", "test_lbl_fixed = y_test_arr_123". Properly paired.
F1	CLOSED	§3.6, five folds, and the disclosure that Naive wins 3 of 5. An unfavourable result reported in full.
F1	CLOSED	§3.7, per-course, and the disclosure that CRAG is not top in 3 of 5 courses.
C3	CLOSED	Table 6 empty column, zero across four systems.
C4 (a)	CLOSED IN CODE	data/gold_qa_pairs.csv and data/gold_system_answers.csv at 710f757, 200 rows each, four answer columns.

The negative results in §3.11 are disclosed properly and worded correctly. CRAG vs Vanilla is called inconclusive rather than absent. That is the right wording and it is not a small thing.

WHAT DID NOT CLOSE

C4 (b) | CLAIMED, NOT PUSHED

This is the serious one. Commit 3f0dcb0's message reads "Add C1-fixed confusion matrices, convergence curves, and reproducible user trust analysis cell". That commit changed three PNG files and nothing else. I searched every commit in the repository for mannwhitneyu, ttest_1samp, nasa, t1x, user-study, and for the reported values 298, 4.1795, 0.8343, 24.3252, 1.8761. The only hits are inside the close-out script's own V6 text at cell 10 L47 and L704.

There is no trust analysis cell. There never was one. user-study.csv is not in the repository. §3.9 reports t=24.3252, U=10145.0, U=11492.5, d=0.3217 and two SDs, none of which any committed code or committed data produces. The script's own V6 line at cell 10 L735 records this item as STILL OPEN, and that line is in the notebook they committed.

CLAIMED, NOT PUSHED: C4 — the Method describes a change absent from 03f6a85. Commit it, then re-run.

NEW OPEN ITEMS, DERIVED FROM THE CODE

X1	ARITY BREAK	cell 7 L73 now returns NINE values. Three call sites still unpack the old arity and one passes the old argument type: cell 8 L6 — 8-value unpack. Raises. cell 13 L5 — 8-value unpack. Raises. Byte-identical duplicate of cell 8. cell 14 L8-11 — 5-value unpack AND passes "X, y" to a
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function whose signature is now
 (questions, labels). Raises twice over.
 cell 10 L456 — the close-out V1d block, same fault.
 As committed, each of these raises.

Two consequences follow, and neither is cosmetic.

§2.2.2 cites "a vocabulary-size self-test comparing a full-dataset fit against a training-only fit" as confirmation of the C1 fix. That self-test is cells 8 and 13. As committed it raises before printing anything.

Table 4 (base rate and per-class recall, seed 42) is produced by V1d at cell 10 L456 and by cell 14. Both raise. Table 4 is captioned "corrected pipeline" and has no executable source in the repository.

X2 | O3 — FIXED IN ONE PLACE ONLY

cell 9 L72 "X = vectorizer.fit_transform(questions)" is still there.
 Cells 14, 15 and 16 consume that global X and that global vectoriser.
 Cell 16 L42 "vectorizer.get_feature_names_out()" reads the full-dataset vocabulary. The leak is fixed in train_model_v2 and left standing in the figure pipeline.

X3 | O2 — STALE ARTEFACTS, BOTH DIRECTIONS

Figures 4, 5 and 6 are unchanged since 2026-09-08 and were produced through the leaked vectoriser. Results §3 says "All figures superseded by this correction are marked accordingly." Three are superseded and unmarked.
 Figures 7, 8 and 12 changed at 3f0dcb0. Cell 14 writes Figures 7 and 8 and raises. No committed code produces those two files.

X4 | SECTION 14 — AN ASSERT THAT FIRED WAS RETARGETED, NOT REPORTED

cell 10 L89 "PINNED = {'scikit-learn': '1.6.1', 'torch': '2.11.0'}".
 The emitted script pinned 1.3.0 and 2.1.2 from requirements.txt. The assert fired on a runtime that had neither. requirements.txt was then edited at 710f757 to match the runtime.
 An assert that fires is a result. The route was to report it as a finding, then decide. §2.5 does disclose the change in prose, which is partial credit, but the guard was made to pass rather than reported as having failed.

X5 | F12 — THE GAP WIDENED UNDER A CORRECTING CAUSE, AND TWO CHANGES SHARE ONE LEDGER ROW

Engineered 0.6214 -> 0.7074. Cohen's d 2.175 -> 3.0005.
 Cause 11 is CORRECTING. The expected direction was stated in advance: the gap shrinks. It grew by 0.086.
 In the same commit, scikit-learn moved 1.3.0 -> 1.6.1 and torch 2.1.2 -> 2.11.0. The ledger carries one cause row for two changes, and the movement runs opposite to that cause's predicted direction.
 This is not an accusation that the fix is wrong. It is that +0.086 currently has no attributable source. One read-only block settles it: run the pre-fix and post-fix pipelines in the SAME environment, 20 seeds each, and print both. Whatever that prints is the result.
 Until then, F12 applies: separate the two, or revert and report 0.6214.

X6 | INTERNAL NUMERIC CONTRADICTION, SETTLED BY ONE PRINTED LINE

§2.4.1 states seed 123 was chosen because its test Macro F1 was "the highest observed".
 §3.2 reports seed 42's engineered Macro F1 as 0.7714.
 §3.8 reports seed 123's as 0.7107.
 Both come from the same engineered_f1s_v2 array. Print max(engineered_f1s_v2) and the seed at that index.

Three smaller code-resolvable points, stated once each. Table 6's SD column carries the between-fold SD beside a per-item mean; the caption discloses it, but it is roughly an order of magnitude smaller than the SD of the distribution the mean describes, so print both. Table 7's confidence intervals have no upper bound on any of the three rows; V1a computes both percentiles. Table 4's column header reads "n (train, 145 total)" over counts summing to 145, which is the full dataset, not the training partition of 101.

One residue on C3, which closed. Zero empty generations, but Table 6 reports 10, 11, 27 and 0 zero-ROUGE items. Twenty-seven non-empty BM25 answers scoring exactly 0.0000 against their gold answer is a generation failure that is not an empty string. Count what those 27 contain.

One artefact point. .gitignore L4 still excludes *.json, so qic_results_corrected.json and gold_rouge_v2.json remain uncommitted. Every classifier number and every ROUGE array still lives only on Drive. Cell 6 L2 loads qic_vectorizer.pkl for the deployed inference path, and that pickle is not in the repository either, so it cannot be established whether the deployed classifier uses the leaked vectoriser or the fixed one.

NULL STATE

NULL STATE: 5 CLAIMED — the manuscript's own MDE analysis supports it.

NOT CONFIRMED — the verdict table was not returned.

EVIDENCE : §3.5, MDE 0.0291 above observed 0.0202, power 0.4436, n=413 required. That wording is correct and I am not disputing it.

One prediction, stated so it can be checked rather than argued. Run cell 10 as committed and V1e will locate `x = vectorizer.fit_transform(questions)` at cell 9 L72, classify it PRE-SPLIT, and set `cause11_confirmed = True`. The script's null-state rule then returns STATE 2, not 5, because a pre-split fit is still in the source. Closing X2 is what moves it to 5.

CLEARANCE CERTIFICATE

NULL CLEARANCE CERTIFICATE

OWNER: GROUP 1 RESULT: CRAG vs Vanilla ROUGE-L COMMIT: 03f6a85

C1 SIGN STABLE ACROSS ≥ 10 SEEDS [FAIL] verdict table not returned
C2 SURVIVES LEAVE-ONE-OUT [FAIL] verdict table not returned
C3 MDE < CLAIMED EFFECT, or INCONCLUSIVE [FAIL] verdict table not returned
C4 BASELINE GOT THE SAME BUDGET, STATED [FAIL] V1f count not returned
C6 TEST SET SCORED ONCE, AFTER FREEZE [FAIL] PERMANENT

SCORE: 0 of 6. NULL STATE: 5 claimed, 2 predicted from the committed source.

C1 to C4 fail only for want of the printed output. The rule is that a condition ticks against a printed number or a commit hash, never against a statement in a manuscript. Paste the verdict table and four of these move in one turn.

C6 is different and it does not move. §2.4.1 discloses that seed 123 was archived for deployment after all 20 runs completed, selected on test Macro F1. That is the test set scored twenty times and then chosen from. No script rescues C6 after the fact. It fails permanently for this result, and the perturbation analysis in §3.8 inherits it, because Table 10 runs on the outcome-selected seed. The disclosure itself is creditable and correctly worded. The consequence is still permanent.

What must go in the limitations is named in one line and no further: the deployment model's seed was selected on test-set outcome, so the test set is not a held-out estimate for that model. Drafting it belongs to your verification report.

NOT MINE

G-A, F3, F4, C2, F7, the withheld claim licence, the §3.3 ablation's standing now that condition C is stated as algebraically identical to condition A, the §2.6 ethics position where 298 participants ran against an approval whose reference number is still pending, and the unresolved presentation note still sitting in the Results file above §3. Writing, judgement and supervision. Nothing above touches them.

VERDICT

ESCALATE — THIS IS A SUPERVISION QUESTION.

ESCALATION PACKET

1. OPEN AT START 8. CLOSED 6: C1 cause 11, C1 cause 4, F8 cause 12, F1 x2,

- C3, C4 gold artefacts. STILL OPEN 1: C4 trust and workload analysis.
NEW 6: X1 arity break, X2 leak in figure pipeline, X3 stale and sourceless figures, X4 retargeted assert, X5 unattributed +0.086, X6 internal contradiction on the highest seed.
2. CONFIRMS RUN: none returned. The corrections ledger in the manuscript is a narrative reconstruction, not the script's computed verdict table.
 3. CLOSE BLOCKS APPLIED: C1 cause 11 at cell 7 L10-25. F8 cause 12 at cell 17 L41-54. Both applied without a returned confirm, ahead of the RUN_CLOSE_BLOCKS gate.
 4. BEFORE AND AFTER, 20 seeds, effective n 101 train / 22 val / 22 test: baseline 0.1975 -> 0.1976; engineered 0.6214 -> 0.7074; p 0.000002 -> 0.000088; d 2.175 -> 3.0005.
RAG arm unchanged and still negative: CRAG vs Naive p=0.554, CRAG vs Vanilla p=0.056, Naive wins 3 of 5 folds, CRAG not top in 3 of 5 courses.
 5. CERTIFICATE: 0 of 6. C6 fails permanently on post-hoc seed selection. C1 to C4 fail pending the verdict table.
 6. THE STUDENT'S OWN READ — required, one of four, committed to and defended: boundary condition, configuration error, modality mismatch, or genuine limit of the engineered component.

NOT CLOSED. STILL OPEN: C4, X1, X2, X3, X4, X5, X6. RUN THE COMMITTED NOTEBOOK TOP TO BOTTOM AND PASTE BACK THE VERDICT TABLE AND EVERY TRACEBACK IT RAISES.

LEDGER — RUN-GROUP_1-20260913
OPEN AT START : 8
CLOSED IN CODE : 6
STILL OPEN : C4, X1, X2, X3, X4, X5, X6
AWAITING PASTE-BACK: the verdict table from cell 10, never returned
NULL STATE : 5 claimed, 2 predicted from source, 0 confirmed
COMPUTE ESTIMATED : unchanged, ceiling still NOT SET

OPEN IN CODE: 8 — C4, X1 to X7. X7 is new and is below.

WHY IT IS ALL TABLES

Three causes, all of them in the code, none of them stylistic.

R01 cited Figure 4, Figure 5, Figure 6 and Figure 7. R02 cites no figure at all. Nine table citations, zero figure citations. The figure references were dropped in the rewrite, not merely unrendered AND UNEXPECTED

The five new analyses in R02 came out of print-only blocks. Tables 7, 8 and 9 and the MDE paragraph in §3.5 were produced by the close-out script's V1a, V1b, V1c and V4. Those blocks print and do not plot, and that is my doing, not the student's. A verify block exists to return evidence for paste-back, so I wrote them to emit aligned text. The student pasted the text into the manuscript, which is the correct thing to do with it. Nothing in that path was ever going to produce a chart.

The old figure pipeline that would have plotted the classifier results is broken or stale. Cell 14, which writes both the confusion matrices and the convergence curves, raises as committed. Cells 15 and 16 still run on the leaked global x and vectorizer from cell 9 L72. So the three cells that could have refreshed the classifier figures either fail or produce pre-fix output.

FIGURE TO SOURCE MAP, AT 03f6a85

FILE	PRODUCER	STATE
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Figure1_System_Architecture.png	none (drawn)	fine, not code
Figure2_Classifier_Architecture.png	none (drawn)	fine, not code
Figure3_Data_Partition.png	none (drawn)	CHECK — shows the split; label now reads 101/22/22
Figure4_PR_ROC_Curves.png	v2 cell 15 L91	LEAKED. global X
Figure5_F1_Distribution.png	v1 cell 59 L72	NO v2 PRODUCER
Figure6_Feature_Importance.png	v2 cell 16 L74	LEAKED. cell 16 L42 reads the full-set vocabulary
Figure7_Confusion_Matrices.png	v2 cell 14 L43	CELL RAISES
Figure8_Convergence_Curves.png	v2 cell 14 L79	CELL RAISES
Figure9_User_Study.png	none, any commit	NO PRODUCER
Figure10_Gold_Eval.png	v2 cell 26 L89	WORKS
Figure11_Latency_Chart.png	v1 cell 48 L87	NO v2 PRODUCER
Figure12_Perturbation_Analysis.png	v2 cell 17 L93	WORKS, post-F8-fix
Figure13_Rag_Evaluation_Chart.png	v2 cell 19 L98	SUPERSEDED. cell 19 L46 states its own reference is CRAG output, not gold

Two of thirteen are current, working and post-fix: Figure 10 and Figure 12.

X7 | FIGURES IN THE CURRENT SET WITH NO PRODUCER IN THE CURRENT PIPELINE

| Figure5_F1_Distribution and Figure11_Latency_Chart were written by the superseded v1 notebook (v1 cell 59, v1 cell 48) and carried forward under new names at b93f075. Figure9_User_Study has no producer in any commit. Three figures in the R02-era set cannot be regenerated from the pipeline that produced the R02 numbers.

ANALYSES WITH NUMBERS AND NO PLOTTING CODE

This is the code gap. It is not a list of what the manuscript should contain.

R02 OUTPUT	PLOTTING CODE AT 03f6a85
Table 3 20-seed baseline vs engineered	NONE in v2 (v1 cell 59 is pre-fix)
Table 4 base rate, per-class recall	cell 14 — raises
Table 5 three-condition ablation	NONE anywhere. cell 18 prints only
Table 6 ROUGE, zero-ROUGE, empties	cell 26 plots the means only; the zero-ROUGE and empty columns have none
Table 7 bootstrap CI, sign flip	NONE. V1a prints only
Table 8 five-fold	NONE. V1b prints only
Table 9 fairness by course	NONE. V4 prints only
Table 10 perturbation	cell 17 — works
Table 11 latency	NONE in v2
§3.5 MDE and achieved power	NONE. cell 27 prints only
§3.9 trust and workload	NONE, and no analysis code either

Eight of eleven result-bearing outputs have no working plot producer in the current pipeline.

Which of these the manuscript carries as a figure, how many figures the target journal wants, and which tables should become figures rather than sit alongside them, are yours. I am not going to select a figure set or write a caption.

SEQUENCING — AND THIS MATTERS MORE THAN THE LIST

Do not have them build these plots yet.

Seven items are open in code and the verdict was ESCALATE. Table 3, Table 4 and Table 5 all rest on the 20-seed arrays, and X5 is unresolved: the +0.086 on the engineered arm has a confounded environment change sitting in the same commit. Table 7, Table 8 and Table 9 rest on arrays whose empty and zero-ROUGE residue is unexplained. Every one of those numbers can still move.

A plot is a rendering of a locked number. Eight plotting blocks built now are eight plotting blocks rebuilt after the verdict table returns, and a figure that disagrees with its own table is worse than no figure.

The order is: verdict table returns, X1 through X7 and C4 close, the numbers stop moving, then one plotting block per locked table, written against the committed arrays rather than against manuscript values. At that point the plot set is cheap, because the arrays already exist as committed CSV.

One item does not wait, because it is a correctness question rather than a presentation one.

Figure3_Data_Partition is a drawn diagram showing the split. The Method now says 101/22/22. If the diagram still shows 102/21/22 it contradicts the corrected text, and that is a code-verified number disagreeing with an artefact in the repository.

LEDGER — RUN-GROUP_1-20260913

OPEN AT START : 8

CLOSED IN CODE : 6

STILL OPEN : C4, X1, X2, X3, X4, X5, X6, X7

AWAITING PASTE-BACK: the verdict table from cell 10, still not returned

NULL STATE : 5 claimed, 2 predicted from source, 0 confirmed

COMPUTE ESTIMATED : unchanged, ceiling still NOT SET

NOT CLOSED. STILL OPEN: C4, X1, X2, X3, X4, X5, X6, X7. PLOTTING BLOCKS ARE BUILT AFTER THE VERDICT TABLE, NOT BEFORE.